The Mechanism of Decay from the Bost–Connes System

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Abstract

Starting from the dynamics of the Bost–Connes state space and the identification of matter within it, we place maintenance of the original particle, transfer into multiparticle final states, and product propagation within one joint action. The parent and product amplitudes evolve together under this action. Distinguishable records retained by the products then specify the probability balance of uninterrupted survival and first decay. Summing first-decay probabilities over all rounds within each channel gives its branching fraction, while weighting the actual elapsed durations by survival probabilities gives the mean lifetime. In the narrow-resonance regime formed by weak coupling and a smooth outgoing spectrum, the same transfer coupling also gives an energy width, relating mean lifetime to spectral width by $\overline{T}_{\mathrm{cont}} = \hbar/\Gamma$.

With explicitly specified effective vertices and response matching, product energy sharing, chiral contractions, and barrier propagation yield continuous spectra, helicity suppression, and exponential lifetime hierarchies, respectively. Long lifetime corresponds to suppressed open exits, while stability requires closure of all relevant exit actions. Why the original particle persists, how it transforms, and what spectra and records its products leave thus become related, calculable results within the same dynamical process.

Keywords: Bost–Connes system; decay dynamics; survival probability; physical records; lifetime and spectral width

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§1 Introduction

The decay of matter is a process in which an original particle or a state of composite matter is transformed through interaction, producing outgoing particles or radiation. A particle can persist and propagate before it decays, while its products continue the subsequent evolution. An excited state of composite matter can also emit particles or radiation and enter a lower-energy state. In that case, it is the original excited state whose survival ends; the composite system persists in its new state.

Dynamics of the Bost–Connes State Space [1] establishes the dynamics of maintenance, exchange, first exit, and re-formation. *The Unification of Matter, Forces, and Spacetime from the Bost–Connes System* [2] further identifies matter and incorporates its internal maintenance, phase evolution, and survival envelope into a complete cycle. On this basis, a physical description of decay must track the transformation of specific particles: which products can form, what amplitudes they acquire, and how these outputs determine the lifetime and decay branches of the original particle and the energy spectra of its products.

Consider a negative muon. Before decaying, it retains its form as matter and continues to propagate. Its dominant decay is $\mu^- \to e^- + \bar{\nu}_e + \nu_\mu$, producing an electron, an electron antineutrino, and a muon neutrino. Together they constitute the multiparticle final state of this decay. Their motions and mutual correlations develop through the interaction. The survival of the original particle and the formation of these products must be compared within the evolution of the same initial preparation.

Locally, the maintenance action continues to act on the original-particle component, while the interaction connects it to allowed product components. The existing amplitudes, phases, and participating correlations all enter the subsequent evolution. Spontaneous muon decay in vacuum involves interactions with the outgoing electron and neutrino degrees of freedom; an incoming collision is not required. A local connection makes conversion possible, while the full process determines the resulting survival and product components.

These local actions jointly determine the complete output. For a normalized pure joint input $|\Psi_{\text{in}}\rangle$, let $\mathbb{U}$ denote the joint evolution operator over an actual time interval. Resolving the output into the original-particle survival component and the different product classes gives

$$\mathbb{U}|\Psi_{\text{in}}\rangle = |\Psi_{\text{surv}}\rangle + \sum_r |\Psi_r^{\text{prod}}\rangle. \tag{1.1}$$

Here $|\Psi_{\text{surv}}\rangle$ is the survival component and $|\Psi_r^{\text{prod}}\rangle$ is the component in product class r; none is separately normalized. Motional components within a product class and their correlations with other degrees of freedom remain in the corresponding joint state. This decomposition specifies the outputs to be tracked. The conversion action determines their amplitudes and phases, while the resolution of product classes and subsequent records determine how the components enter observable statistics.

The original particle, its products, and the other actual participants jointly account for energy, momentum, and conserved charges. Available energy restricts the possible final states, and interaction matrix elements further determine which allowed states acquire nonzero amplitudes. In muon decay, the electron and the two

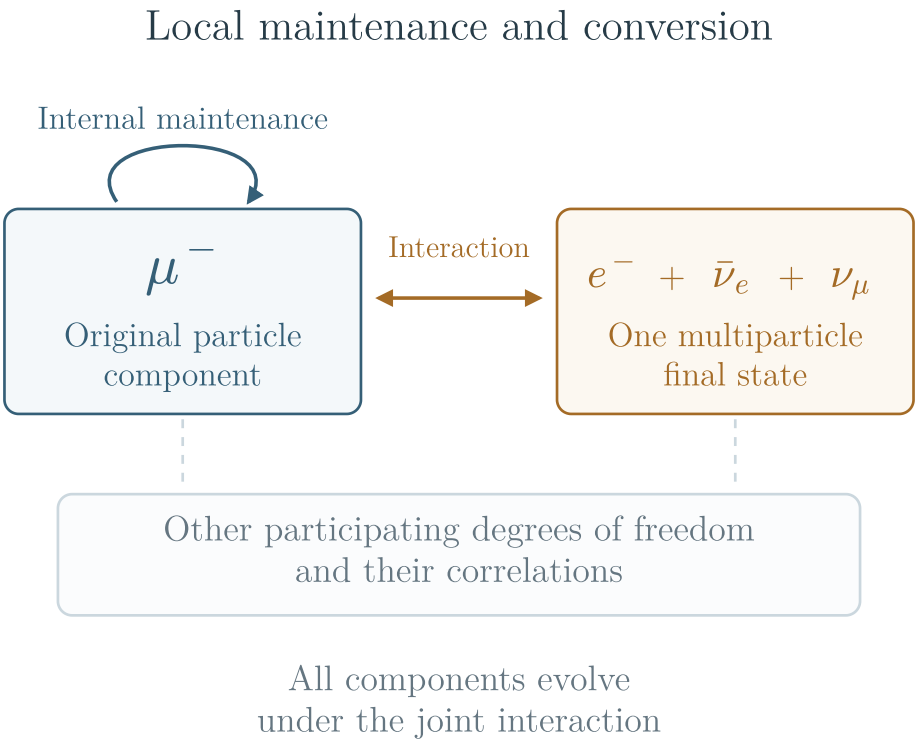


Figure 1-1. Local action of decay. In the dominant muon decay channel, the original-particle component participates in internal maintenance and is connected to a final state containing an electron, an electron antineutrino, and a muon neutrino. The bidirectional line represents an amplitude connection in the joint interaction; the initial preparation and subsequent evolution determine the actual outgoing state.

neutrinos must be treated together in a three-body final state. The electron energy distribution and the total decay rate follow from the same final-state amplitudes.

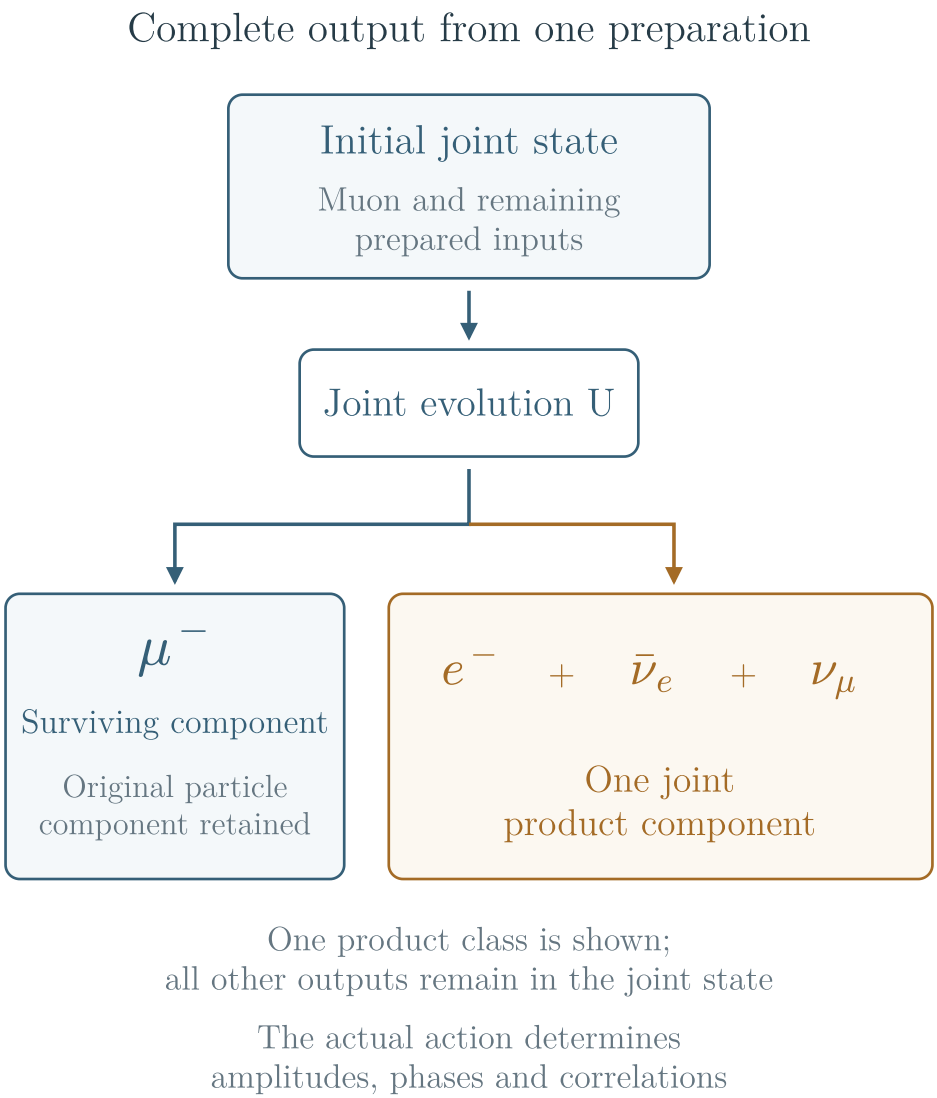


Figure 1-2. Complete output of the same action. The upper part shows an initial preparation and its joint evolution; below it are the original-particle survival component and one class of multiparticle products. The electron and both neutrinos belong to a single final state. The branches in the diagram represent components of the joint output.

The complete output continues to evolve, so a single action becomes part of a continuous history. Internal maintenance and spatial propagation continue along the component in which the original particle has not yet decayed. Components transferred into the product space undergo the propagation and interactions of their constituent products. First decay ends the uninterrupted survival of the original particle. If a particle of the same kind forms again, its occupation continues within the same joint evolution but is counted separately from uninterrupted survival before first decay [1,2].

Subsequent interactions of the products can preserve distinctions associated with this conversion. The electron's response in detecting matter, states carried by the

other products, and their correlations jointly determine which decay information can be distinguished. Physical interactions form the record; readout accesses distinctions already stored in it. Its resolving power depends on the output actually retained and maintained [1]. Conversion of the original particle, record formation, and reception at a distant detector can occur at different stages. Their times must be determined from the actual durations of propagation and interaction.

Repeating the same preparation allows the shares that survive uninterrupted to different times to be compared, and yields statistics of first-decay times, product classes, energies, and directions. Lifetime describes how long the original state persists, branching fractions describe the proportions of different products, and spectra describe energy sharing within each product class. Actual interactions and complete time accounting relate these quantities.

This picture also covers the decay of other particles and composite matter. Weak interactions can change particle species. Strong and electromagnetic interactions can cause unstable particles to decay or excited composite systems to relax. Constituents of a nucleus can also separate into outgoing products after traversing a barrier. Each process has its own interaction strength, final-state structure, and opening conditions. Survival of the original state, formation of products, and their subsequent records provide a common description for comparing these processes.

We address three related questions:

1. How does matter transform while it persists, and how do specific decay products form?
2. How do these products preserve distinguishable decay records that specify first events and final-state branches?
3. How do interactions, final-state structure, and total elapsed duration determine lifetimes, spectra, and the stability of matter?

Answering these questions first requires following the same particle preparation and calculating how the interaction retains the original-particle component while forming specific multiparticle outputs.

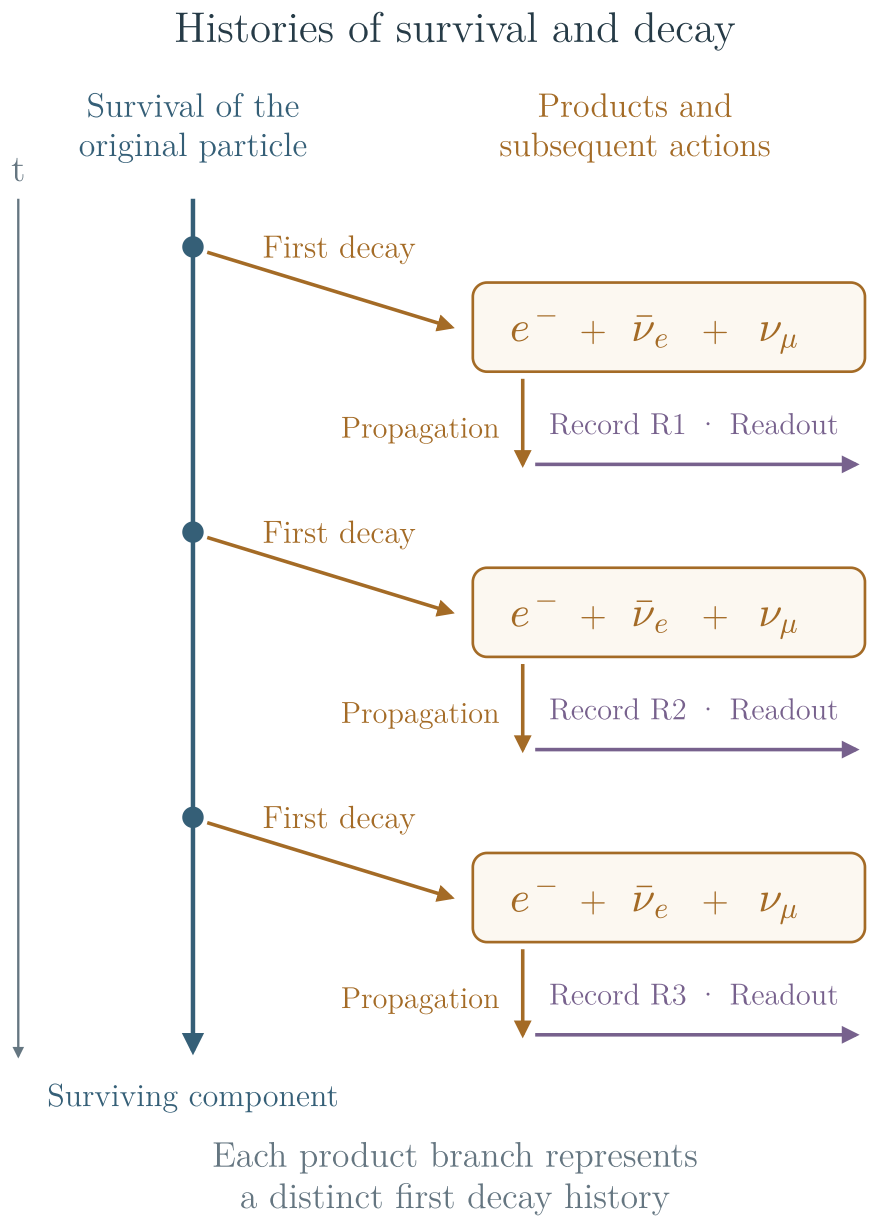


Figure 1-3. Continuous histories of survival, decay, and products. Actual elapsed duration increases downward. The continuing component on the left represents survival of the original particle; branches to the right show products emitted at different stages. Each product output continues to evolve. Distinctions retained later correspond to distinguishable outcomes through readout arrows marked with the record R. The diagram organizes different first-decay histories, rather than repeated decays of one original particle. Cycle counts, record counts, and elapsed durations are tracked separately.

§2 The Decay of Matter

Before decay, the original particle continues to be maintained and to propagate; after conversion, the resulting particles together constitute a new state of matter. Calculating this change requires a single action that incorporates the internal connection of the original particle, the formation of additional products, and their motion. Using muon decay as an example, we construct an effective realization connecting one parent state to multiparticle final states in the established matter representation, and follow its complete amplitude evolution.

§2.1 Original Particle and Products

Internal maintenance and propagation sustain an already formed particle [2], while conversion connects different particle compositions. The state space used in the calculation must therefore contain both the original particle and the products under consideration.

First choose a finite window of motional modes in a finite volume, and fix a momentum and spin preparation of the original particle. Denote the normalized state containing this preparation and the required auxiliary inputs by $|i\rangle$, and call it the selected parent state. For muon decay, the product space is spanned by joint states of the electron, electron antineutrino, and muon neutrino. Any additional outputs are included according to the chosen action. An orthonormal final-state basis $\{|f_\alpha\rangle\}$ satisfies

$$\langle i|f_\alpha\rangle = 0, \qquad \langle f_\alpha|f_\beta\rangle = \delta_{\alpha\beta}. \tag{2.1}$$

The label α resolves the products' motion, spin, and any other degrees of freedom still participating in the interaction. The electron and both neutrinos belong to one final state; changing the motional component of any one of them can yield a different α. The finite mode window specifies the normalization and summation convention used here. A continuum final-state limit can then be considered within the range of validity of the interaction.

Let $P_i = |i\rangle\langle i|$ and $P_F = \sum_\alpha |f_\alpha\rangle\langle f_\alpha|$ project onto this parent state and the product space, respectively. In the selected effective model, the complete generator is

$$\boxed{\begin{aligned} &H = E_i P_i + H_F + B + B^\dagger, \\ &B = P_F H P_i, \qquad H_F = P_F H P_F. \end{aligned}} \tag{2.2}$$

Here E_i is the uncoupled energy of the parent state, H_F governs product propagation and the internal interactions retained, and B connects the parent state to the products. The reverse connection $B^\dagger$ makes the joint generator self-adjoint. Additional parent states or outputs enlarge these blocks. We set $\hbar = 1$ below; time measures the actual elapsed duration of the action generated by this operator.

This block structure is consistent with the self-adjoint exchange connecting inputs to complete outputs in the common dynamics [2]. A particular decay also requires the particle composition and matrix elements of B. They must connect the internal matter representations and form the corresponding motional final states.

§2.2 Local Conversion

Internal components of particle species can be connected through a common intermediate state. Take the zero-element state $|z\rangle$ in formation space and normalized nontrivial directions $|\psi_j\rangle$ on the unit shell. Define the absorption action $F_j = |z\rangle\langle\psi_j|$, with formation action $F_k^\dagger$. Then [2]

$$F_k^\dagger F_j = |\psi_k\rangle\langle\psi_j|. \tag{2.3}$$

The operator on the right transfers the component in direction j into direction k. The zero element and directions here belong to formation space; separate position degrees of freedom carry the particle's spatial position.

This intermediate connection can be calculated directly on the $L^2(\mathbb{Z}/p\mathbb{Z})$ section. Write $z = \delta_0$, $P_z = |z\rangle\langle z|$, and let the additive translation of formation addresses be $(\tau_a f)(x) = f(x + a)$. For a nonzero address a,

$$\begin{aligned} C_a &= [P_z, \tau_a] \\ &= |z\rangle\langle\delta_a| - |\delta_{-a}\rangle\langle z|. \end{aligned} \tag{2.4}$$

Normalized unit-shell directions satisfy $\psi_j(0) = 0$. The first action sends the component at address a to the zero element; the second connects the zero element to another address:

$$\begin{aligned} C_a|\psi_j\rangle &= \psi_j(a)|z\rangle, \\ C_bC_a|\psi_j\rangle &= -\psi_j(a)|\delta_{-b}\rangle. \end{aligned} \tag{2.5}$$

The one-step and two-step matrix elements between directions are therefore

$$\begin{aligned} \langle\psi_k|C_a|\psi_j\rangle &= 0, \\ \langle\psi_k|C_bC_a|\psi_j\rangle &= -\overline{\psi_k(-b)}\psi_j(a). \end{aligned} \tag{2.6}$$

With $N = p-1$, the directional waveform on the unit shell is $\psi_j(a) = \chi_j(a)/\sqrt{N}$, so the modulus of the two-step matrix element is $1/N$. This is the result for the selected local transfer. Complete emission also includes intermediate propagation, coupling strength, product motion, and overlaps of the other outputs. In general, C_a is not itself a self-adjoint generator; actual exchange must be constructed with the corresponding adjoint and the complete participating space.

For muon decay, the internal connection from a charged lepton to a neutrino can be represented by $F_3^\dagger F_2$. It changes the species component on the original-particle line, while the same emission action must also form the electron and electron antineutrino. The zero-element intermediate state also permits coherent retrieval [1,2]. Subsequent evolution determines whether the transferred component becomes a persistently outgoing product.

§2.3 Formation of Multiparticle Products

Single-particle conversion and the formation of additional particles must be implemented in the same operator. Following the identification of fermionic matter in [2], we organize multiparticle modes in antisymmetric Fock space and denote the vacuum of these particle modes by $|\mathrm{vac}\rangle$. It is defined separately from the formation-space

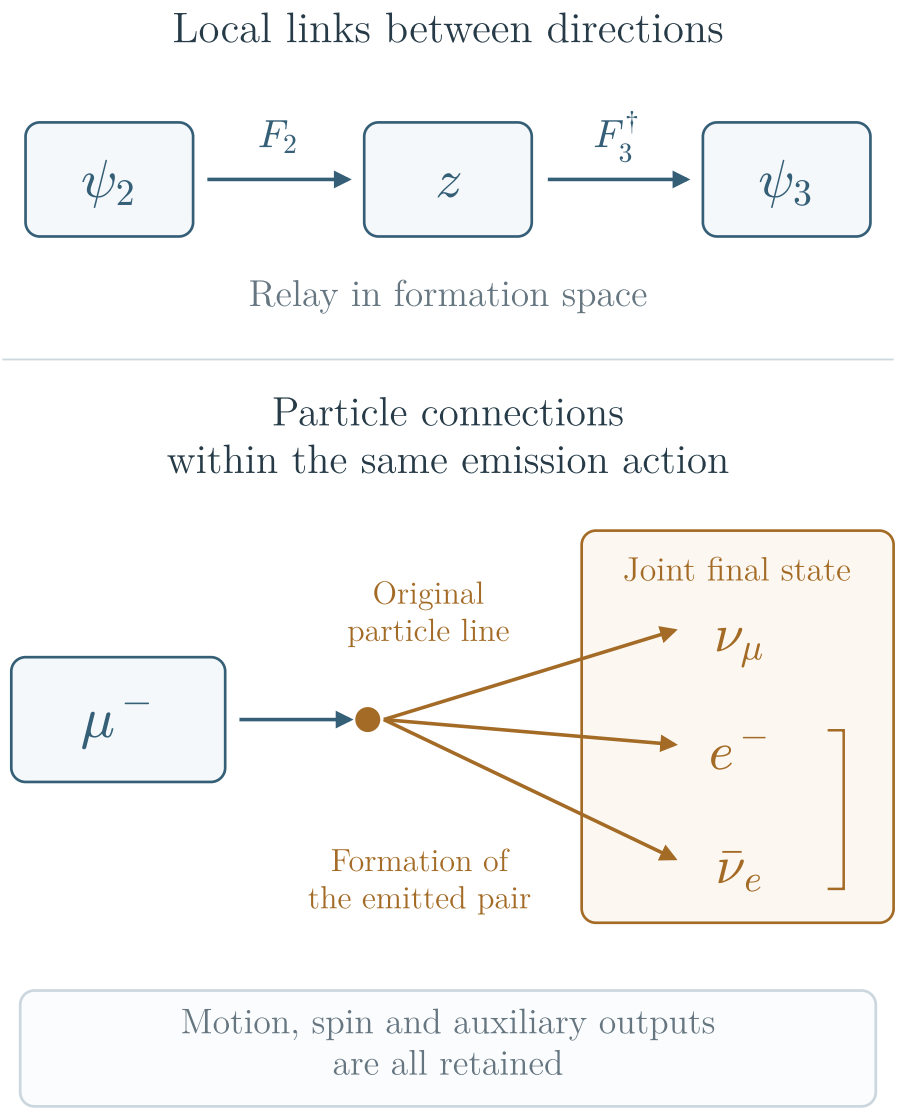


Figure 2-1. Internal conversion and multiparticle formation. The upper part shows a local action connecting directions through the zero element in formation space. The lower part shows the full particle connection for muon decay. The muon neutrino and the electron–electron-antineutrino pair together form the final state.

zero element $|z\rangle$. Let b_μ, b_e, and b_{ν_μ} be the annihilation operators of the corresponding modes, and d_{ν_e} the annihilation operator of the electron-antineutrino mode.

Writing the particle part first and suppressing momentum and spin indices, the minimal emission term is

$$\mathcal{C}_\alpha = b_e^\dagger d_{\nu_e}^\dagger b_{\nu_\mu}^\dagger b_\mu. \tag{2.7}$$

The order of the creation operators is fixed, retaining the signs from fermionic reordering. If the auxiliary input $|a_0\rangle$ is connected to $|a_\alpha\rangle$, the complete term also contains $R_\alpha = |a_\alpha\rangle\langle a_0|$, giving $\mathcal{C}_\alpha \otimes R_\alpha$. The label α resolves both particle modes and auxiliary outputs. For an input containing a single muon, the particle anticommutation relations give

$$\begin{aligned} &\mathcal{C}_\alpha b_\mu^\dagger |\mathrm{vac}\rangle \\ &\qquad = b_e^\dagger d_{\nu_e}^\dagger b_{\nu_\mu}^\dagger |\mathrm{vac}\rangle. \end{aligned} \tag{2.8}$$

The original particle is annihilated and all three products are formed in the same action. The labels ν_μ and $\bar{\nu}_e$ denote the neutrino states coupled at the charged-current endpoints; their subsequent propagation is resolved under the actual neutrino mass action. Compatibility of the neutrino species direction with its conjugate does not remove the need to distinguish the spin and endpoint states within the complete final state [2].

Within the selected one-parent-state and three-body-final-state subspace, assigning coefficients to these emission terms and their adjoints gives

$$\boxed{\begin{aligned} B &= \sum_\alpha g_\alpha |f_\alpha\rangle\langle i|, \\ V_\mu &= B + B^\dagger. \end{aligned}} \tag{2.9}$$

This is the minimal effective charged-current connection adopted here. The coefficient g_α has units of energy and includes the vertex's motion, spin, auxiliary overlaps, and strength normalization. Chapter 5 adopts a low-energy contact form and matches its strength to evaluate these coefficients.

This vertex connects μ^- to ν_μ and creates one $e^-\bar{\nu}_e$ pair. The total fermion number changes from one to three, preserving fermion parity. Exactly one emitted pair specifies the scope of this minimal process.

§2.4 Decay Channels and Conservation

Conservation relations of the complete action constrain the particle composition of the final state. Charge is added over the representations of the particles and actual participants. If Q is the joint charge operator and the initial and final states have definite charges, then

$$\begin{aligned} &[Q, V_\mu] = 0 \\ \Longrightarrow \quad &(q_\alpha - q_i) g_\alpha = 0. \end{aligned} \tag{2.10}$$

The initial charge in muon decay is $-e$, as is the total charge of the electron and two neutrinos. Along the original-particle line, $\mu^- \to \nu_\mu$ increases the charge by e,

while the emitted $e^-\bar{\nu}_e$ pair carries $-e$. Both contributions balance within the same vertex.

The group labels at the species level can also be checked. We use the six-direction convention of [2], with indices added modulo 6. The charged-lepton direction is χ_2, and the neutrino direction is χ_3, giving

$$\begin{aligned} \chi_3\chi_2\chi_3 &= \chi_2, \\ \chi_3\chi_2^{-1} &= \chi_1, \qquad \chi_2\chi_3 = \chi_5, \\ \chi_1\chi_5 &= 1. \end{aligned} \tag{2.11}$$

The label change on the original-particle line is compensated by the emitted pair, making the total initial and final species labels compatible. Physical charge, color, and spin are checked in their respective complete representations [2]. The actual vertex connects generation branches; this example uses the muon and its corresponding neutrino endpoint.

Applying the same charged-current structure of one fermion line and one emitted pair to leptons and quarks gives the representative connections below. The indices j, k, l, m label the corresponding generations; the actual vertex supplies the nonzero coefficients.

Original-particle line	Emitted pair	Representative final state
Down-type to up-type quark	Lepton pair	$d_j \to u_k + \ell^- + \bar{\nu}_\ell$
Down-type to up-type quark	Quark pair	$d_j \to u_k + d_l + \bar{u}_m$
Charged lepton to neutrino	Lepton pair	$\ell_j^- \to \nu_j + \ell_k^- + \bar{\nu}_k$
Charged lepton to neutrino	Quark pair	$\ell_j^- \to \nu_j + d_l + \bar{u}_m$

Table 2-1. Representative connections of the minimal charged-current vertex.

These connections list particle compositions accommodated by the selected minimal vertex. Which channels actually open also depends on their matrix elements, available energy, and the internal states of composite matter. The quark rows describe connections participating in weak interactions inside hadrons; experimentally outgoing states must still be assembled into allowed hadrons and other particles.

Spatial translations and rotations are likewise checked in the actual joint representation. If the action commutes with the total momentum $\mathbf{P}$, nonzero matrix elements between definite-momentum states satisfy

$$\begin{aligned} &[\mathbf{P}, V_\mu] = 0 \\ \Longrightarrow \quad &\mathbf{p}_i = \mathbf{p}_e + \mathbf{p}_{\bar{\nu}_e} \\ &\qquad + \mathbf{p}_{\nu_\mu} + \Delta\mathbf{p}_{\text{aux}}. \end{aligned} \tag{2.12}$$

Here $\Delta\mathbf{p}_{\text{aux}}$ is the net momentum change of the actual auxiliaries; it vanishes in an isolated three-body process. Angular momentum includes both spin and orbital parts, and the full vertex must respect their joint rotations. Conservation restricts where nonzero matrix elements can occur; the interaction determines the amplitudes at those allowed positions.

§2.5 Amplitude Transfer between Parent and Products

Once the product composition is specified, we must calculate the amplitudes generated in those product components and their shares relative to the initial preparation. Change the product-space basis to an orthonormal eigenbasis of H_F, satisfying $H_F|f_\alpha\rangle = E_\alpha|f_\alpha\rangle$, and retain g_α for the matrix elements of the same emission action in this basis. This basis change also includes the expansion of internal states such as those of neutrinos.

In the single-parent-state model, the state is

$$\begin{gathered}|\Psi(t)\rangle = A(t)|i\rangle + \sum_\alpha B_\alpha(t)|f_\alpha\rangle, \\ A(0) = 1, \qquad B_\alpha(0) = 0.\end{gathered} \tag{2.13}$$

Applying the generator and projecting onto the parent and each final state yields

$$\boxed{\begin{aligned} i\dot{A} &= E_i A + \sum_\alpha \overline{g_\alpha} B_\alpha, \\ i\dot{B}_\alpha &= E_\alpha B_\alpha + g_\alpha A. \end{aligned}} \tag{2.14}$$

The first equation retains both the phase evolution of the original particle and contributions returning from the products. The second describes products propagating while continuing to receive contributions from the original-particle component. Define the directed share current between the parent and final state α by

$$\mathcal{I}_\alpha(t) = 2\,\mathrm{Im}\left[\overline{B_\alpha(t)} g_\alpha A(t)\right] \tag{2.15}$$

Differentiating the squared moduli of the amplitudes then gives

$$\begin{gathered}\frac{d|A|^2}{dt} = -\sum_\alpha \mathcal{I}_\alpha, \qquad \frac{d|B_\alpha|^2}{dt} = \mathcal{I}_\alpha, \\ |A|^2 + \sum_\alpha |B_\alpha|^2 = 1.\end{gathered} \tag{2.16}$$

The share acquired by the final states comes from the change in the parent share under the same action. The sign of the current depends on the amplitudes and phases, and the finite model permits return. The quantities $|A(t)|^2$ and $\sum_\alpha |B_\alpha(t)|^2$ are the instantaneous parent and product occupations. Uninterrupted survival before first decay must be counted separately along actual records.

The product amplitude can also be resolved by formation time. Multiplying the second amplitude equation by $e^{iE_\alpha t}$ and integrating gives

$$\boxed{B_\alpha(t) = -ig_\alpha \int_0^t e^{-iE_\alpha(t-u)} A(u)\, du.} \tag{2.17}$$

A contribution formed at time u evolves to t with the product propagation phase. Contributions from different times add within the same final-state amplitude. Substituting this expression into the parent equation and writing $a(t) = e^{iE_i t}A(t)$ gives

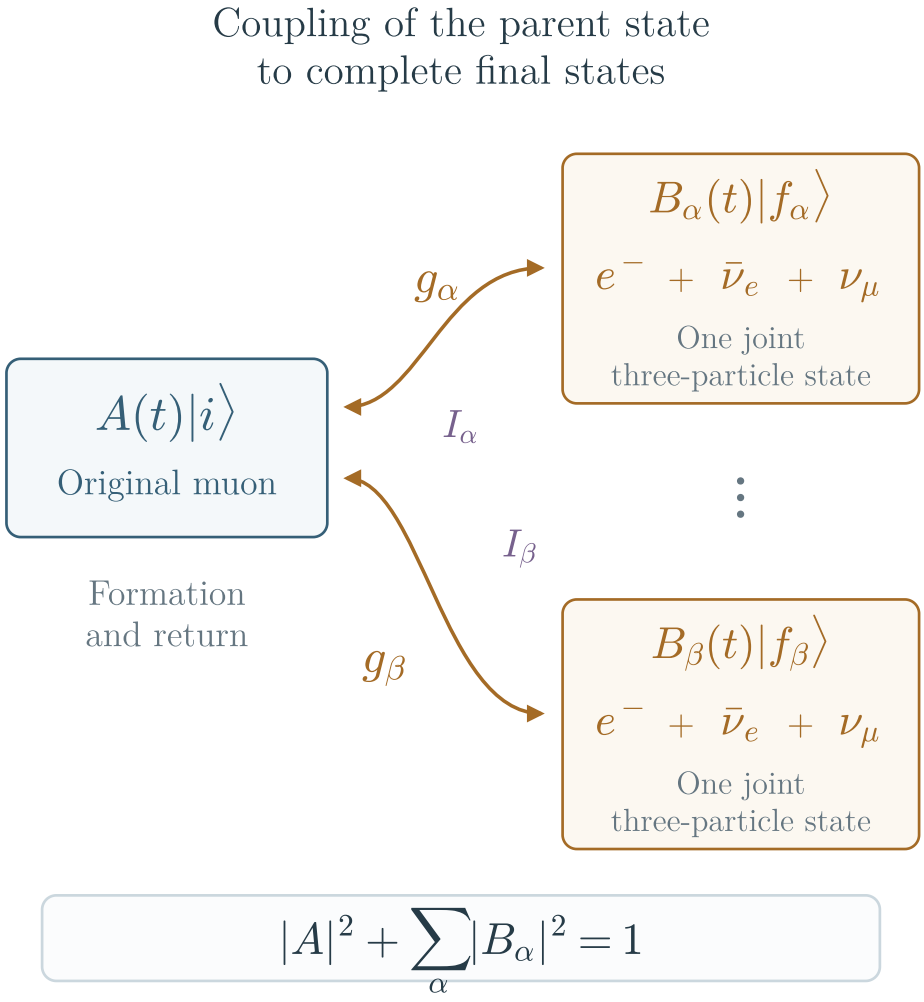


Figure 2-2. Amplitude transfer between parent and products. The complete coefficients g_α connect one selected parent state to different three-body final states, each containing all three products and their correlations. Amplitudes and phases at both ends determine the directed share current, while all components together preserve normalization.

$$\begin{aligned} \dot{a}(t) &= -\int_0^t K(t-u)a(u)\,du, \\ K(s) &= \sum_\alpha |g_\alpha|^2 e^{-i(E_\alpha - E_i)s}. \end{aligned} \tag{2.18}$$

The memory kernel K is built from the final-state frequencies and couplings, retaining both product formation and coherent return in the parent evolution.

While conversion remains weak and parent depletion is negligible, take $A(u) \simeq e^{-iE_i u}$. Integration yields

$$\begin{aligned} B_\alpha^{(1)}(t) = -ig_\alpha t\, e^{-i(E_i+E_\alpha)t/2} \\ \times \operatorname{sinc}\left[\frac{(E_\alpha - E_i)t}{2}\right], \\ \operatorname{sinc} x = \frac{\sin x}{x}. \end{aligned} \tag{2.19}$$

Nearly energy-matched components can accumulate, whereas large mismatches suppress them through phase variation between formation times. The time-independent total generator H conserves total energy. During a finite interval in which the interaction remains active, the uncoupled energies E_i and E_α need not match term by term; interaction energy also enters the complete balance. Energy selection and rates for continuous final states over appropriate time intervals follow from this integral.

For freely outgoing products after separation from the interaction, an energy-matched description of a parent particle with a narrow energy spread balances initial and final energies together with any actual auxiliaries. In the calibrated long-wavelength regime of relativistic propagation, a necessary condition for an isolated parent at rest to have available three-body phase space is

$$m_A > m_1 + m_2 + m_3. \tag{2.20}$$

An open energy threshold must still be accompanied by a nonzero matrix element. Thresholds for composite matter use the energies of the complete initial and final composite states, including binding and internal excitation.

§2.6 Subsequent Evolution of the Products

Emission forms a multiparticle state with amplitudes and correlations. After the products leave the local interaction region, and over intervals in which their mutual interactions are negligible, the propagation generator is additive over the particles. Hence

$$\begin{aligned} H_F &= H_e + H_{\bar{\nu}_e} \\ &\quad + H_{\nu_\mu} + H_{\text{aux}}, \\ U_F(t) &= U_e(t) \otimes U_{\bar{\nu}_e}(t) \\ &\quad \otimes U_{\nu_\mu}(t) \otimes U_{\text{aux}}(t). \end{aligned} \tag{2.21}$$

The neutrino propagation operators act on the full mass-state spaces of the corresponding products; their subscripts retain the production-endpoint labels. Factorization of the propagation action does not require factorization of the product

state. Momentum, spin, and internal correlations established at emission remain in the joint state. When a product encounters detecting matter, this actual state determines the subsequent response. If the products continue to interact, they evolve under the corresponding complete H_F.

For an emission interval $t_1 - t_0$ followed by propagation to t_2, the product component is

$$|\Psi_F(t_2)\rangle = U_F(t_2 - t_1)P_F \times e^{-iH(t_1-t_0)}|i\rangle. \tag{2.22}$$

The projector P_F selects the output component being tracked; the parent and other outputs remain in the complete history. Selecting a component by a projector does not itself add a detection interaction. Emission, propagation, actual return, and processing follow their physical schedule, from which the total duration of a complete cycle is accumulated [1,2].

When products interact with other matter, distinctions between conversion processes can remain in the new outputs. These subsequent actions determine which distinctions persist and which can still disappear through coherent return. Decay records must be established from the specific products and their joint state before the statistics of first decay, final-state branches, and cascades can be specified.

§3 Records of Decay

Decay products continue to propagate and interact with surrounding matter. Distinctions left by these interactions can identify whether the original particle has transformed, which product class has formed, and the interval in which it occurred. The amplitudes obtained in Chapter 2 thereby enter actual records, giving the same original-particle preparation a traceable decay history.

§3.1 Records in the Products

The electron and both neutrinos in muon decay jointly carry the state after emission. The electron produces a response in detecting matter while the other products propagate, and their correlations remain part of the complete output. Identifying a decay does not require direct detection of every product, but the chosen response must distinguish decay, survival of the original particle, and background processes for the given initial preparation.

We use three criteria for a complete record: distinctions must be identifiable, actually remain after the complete action, and persist throughout readout [1,2]. Let h label process components determined by the actual preparation and evolution, $r = f(h)$ the class to be recorded, and ω_h the normalized conditional state of the record carrier. The actual pointer projections Π_r^{rec} satisfy

$$\Pi_r^{\mathrm{rec}}\Pi_s^{\mathrm{rec}} = \delta_{rs}\Pi_r^{\mathrm{rec}}, \qquad \sum_r \Pi_r^{\mathrm{rec}} = I. \tag{3.1}$$

The outcomes include no response. Error-free identification of different classes requires their conditional states to lie in the corresponding pointer subspaces:

$$\boxed{\Pi_r^{\mathrm{rec}}\omega_h = \delta_{r,f(h)}\omega_h.} \tag{3.2}$$

The output used here is taken after emission, propagation, and any actual retrieval. An intermediate distinction meets the retention criterion only if it still has nonzero weight in that output. Formation of the electron and neutrinos does not by itself mean that a particular apparatus has recorded all their momenta and spins.

Let the record form at t_{form} and be read at t_{read}. Extend the pointer projection by the identity on the remaining degrees of freedom, writing it as $\widehat{\Pi}_r^{\mathrm{rec}}$. If the actual joint evolution U_{post} during storage satisfies

$$U_{\mathrm{post}}^{\dagger}\widehat{\Pi}_r^{\mathrm{rec}}U_{\mathrm{post}} = \widehat{\Pi}_r^{\mathrm{rec}}, \tag{3.3}$$

the share associated with each label is preserved. This can be realized when the carrier leaves the emission region and no longer participates in coherent retrieval into the original process, while storage preserves the relevant pointer. The distinction remains in the joint output regardless of whether anyone reads it. Readout identifies it and assigns physical readings through the actual pointer calibration.

A partially distinguishable feature can still provide partial information, but does not give an error-free classification of all processes. Whether coherence remains is determined by the actual overlaps of all relevant outputs.

§3.2 Record Resolution

The same muon decay can be reported by particle composition, or further grouped by electron energy interval or outgoing direction. The label α in Chapter 2 resolves a joint final-state basis; an actual record r generally retains only some of those distinctions. Unresolved motion and internal states within the same record class still participate in subsequent evolution.

For processes leading to the same normalized matter final state $|\phi\rangle$, write their amplitudes as a_h and the normalized conditional states of the remaining outputs as $|e_h\rangle$. Together they form

$$|\Omega_\phi\rangle = |\phi\rangle \otimes \sum_h a_h|e_h\rangle. \tag{3.4}$$

With normalization relative to the same input, the share of the selected final state is [1,2]

$$\boxed{w_\phi = \sum_{h,k} a_h\overline{a_k}\langle e_k|e_h\rangle.} \tag{3.5}$$

When the remaining outputs are identical, the processes add at the amplitude level. When they are pairwise orthogonal, cross terms vanish and the share is $\sum_h |a_h|^2$. Partial overlap retains cross terms weighted by the actual inner products.

The formation addresses in Chapter 2 provide an explicit calculation. Again take $N = p - 1$ and $\psi_j(a) = \chi_j(a)/\sqrt{N}$. A local absorption action that distinguishes every nonzero address with equal strength can be written as

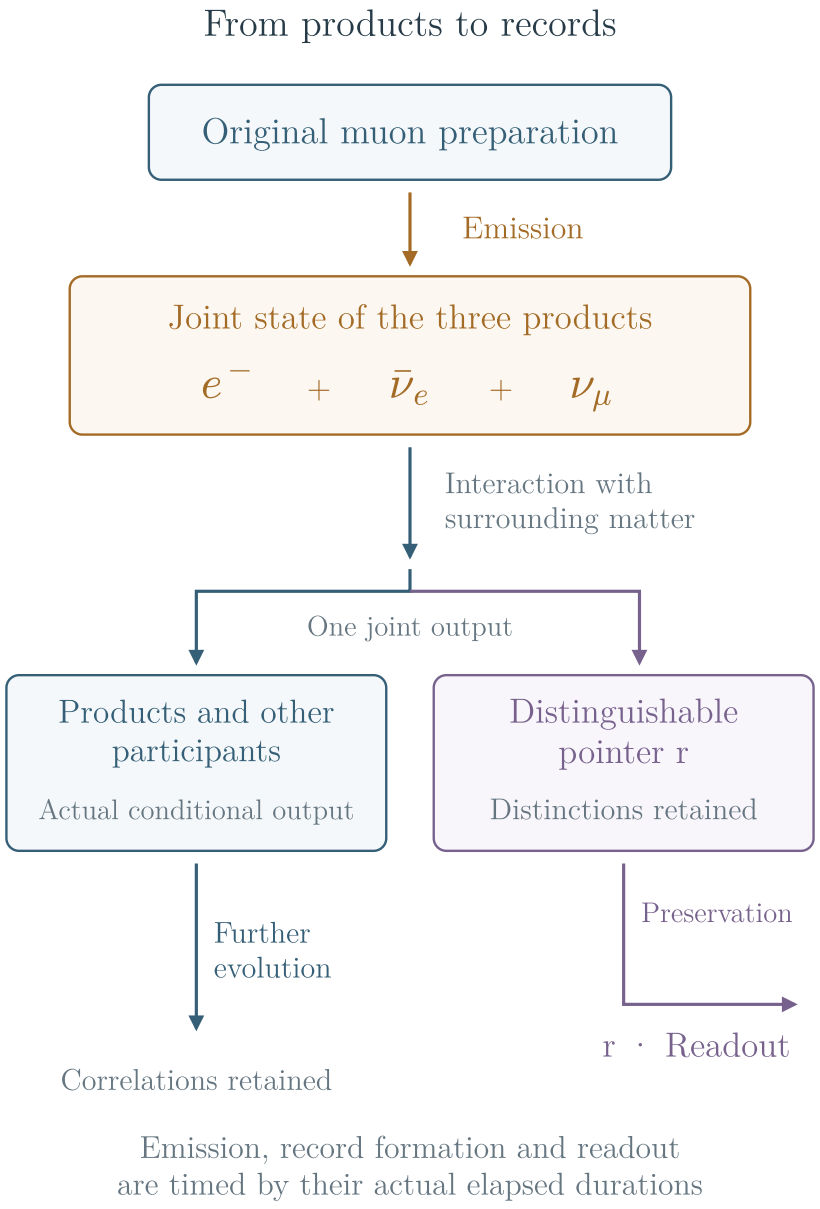


Figure 3-1. Product formation, record storage, and readout. The electron and two neutrinos first form a joint final state. Subsequent interactions with other matter can leave distinguishable records. The two lower parts together constitute a joint output: evolving products and correlations are retained, and the record label is obtained along a readout arrow after storage. Product emission, record formation, and distant readout have their respective elapsed durations.

$$\begin{aligned} \mathsf{A}_{\mathrm{pt}} &= \kappa \sum_{a\neq 0} |z, R_a\rangle\langle \delta_a|, \\ \langle R_b|R_a\rangle &= \delta_{ab}, \qquad |\kappa| \leq 1. \end{aligned} \tag{3.6}$$

Here κ is the dimensionless transfer amplitude of this action; the unabsorbed component is included in the complete output. Orthogonal labels are left for different addresses, giving

$$\begin{aligned} w_j(a) &= |\kappa|^2 |\psi_j(a)|^2 = \frac{|\kappa|^2}{N}, \\ \sum_{a\neq 0} w_j(a) &= |\kappa|^2. \end{aligned} \tag{3.7}$$

The total local share in this equal-strength realization is independent of direction j. It determines only this absorption factor; the complete g_α of Chapter 2 also contains emission, motion, spin, and the actions on other outputs.

If already recorded addresses a and $-a$ are merely combined into a report class d, counting each pair once, the corresponding address projection is

$$\begin{aligned} E_d &= |\delta_a\rangle\langle\delta_a| + |\delta_{-a}\rangle\langle\delta_{-a}|, \\ w_j(d) &= |\kappa|^2 \langle\psi_j|E_d|\psi_j\rangle = \frac{2|\kappa|^2}{N}. \end{aligned} \tag{3.8}$$

This share is the same for odd and even directions. In contrast, if the actual action combines the two addresses in phase into one outlet, the selected input combination is

$$|D_d^+\rangle = \frac{|\delta_a\rangle + |\delta_{-a}\rangle}{\sqrt{2}}, \tag{3.9}$$

and

$$\begin{aligned} \langle D_d^+|\psi_j\rangle &= \frac{\chi_j(a)[1+\chi_j(-1)]}{\sqrt{2N}}, \\ \sum_d |\langle D_d^+|\psi_j\rangle|^2 &= \frac{1+\chi_j(-1)}{2}. \end{aligned} \tag{3.10}$$

Even directions pass through this in-phase outlet, while odd directions cancel there. The orthogonal out-of-phase combination and other outputs must still be included in the complete action. Similarly, when all addresses enter a single coherent outlet with equal coefficients, nontrivial directions satisfy $\sum_a \chi_j(a) = 0$. These cancellations arise from specific coherent actions; combining already recorded outcomes only adds their shares. Throughout this calculation, a denotes a formation address. The particle's spatial position is resolved separately through its motional degrees of freedom and the corresponding pointers.

§3.3 Conditional Outputs of the Products

A record specifies both an outcome and the state of matter that continues to evolve after it. Include the parent particle, products still participating in the interaction, and old auxiliaries in S. Let D denote record degrees of freedom independently prepared for this interval and no longer coupled back after the action. With preparation η_D and actual joint evolution $\mathbb{U}$ including emission, propagation, and record formation,

$$\Omega^{\text{out}}(\rho) = \mathbb{U}(\rho \otimes \eta_D)\mathbb{U}^\dagger. \tag{3.11}$$

Selecting the actual outcome with $\widehat{\Pi}_r = I_S \otimes \Pi_r^{\text{rec}}$ and tracing out D gives the conditional map and its probability [1]:

$$\boxed{\begin{aligned} \mathcal{I}_r(\rho) &= \operatorname{Tr}_D\left[\widehat{\Pi}_r \Omega^{\text{out}}(\rho)\widehat{\Pi}_r\right], \\ p_r &= \operatorname{Tr}\mathcal{I}_r(\rho). \end{aligned}} \tag{3.12}$$

The maps for all outcomes form a quantum instrument. For a complete set of outcomes, $\mathcal{F} = \sum_r \mathcal{I}_r$ is trace preserving and $\sum_r p_r = 1$.

After the emission interval has ended and the product components have left the emission region, suppose an ideal readout of the complete output groups orthogonal final states by $r = f(\alpha)$ and fully distinguishes the parent from the products. Normalized to the original parent-particle preparation, its probabilities are

$$p_r(t) = \sum_{\alpha: f(\alpha)=r} |B_\alpha(t)|^2. \tag{3.13}$$

This relates the final-state amplitudes of Chapter 2 to this ideal readout. Combining several already distinguished outcomes into $\mathscr{A}$ gives

$$\mathcal{I}_{\mathscr{A}} = \sum_{r\in\mathscr{A}} \mathcal{I}_r, \qquad p_{\mathscr{A}} = \sum_{r\in\mathscr{A}} p_r. \tag{3.14}$$

For $p_r > 0$, the normalized conditional output is

$$\rho_r = \frac{\mathcal{I}_r(\rho)}{p_r}. \tag{3.15}$$

In an ideal readout that retains the electron, identifying an energy interval leaves its within-interval distribution, both neutrino states, and their correlations with the electron in ρ_r. After absorptive detection, the response of the detecting matter and its correlations with other products carry the evolution forward. Subsequent propagation and interactions begin from the actual conditional output. When no particular outcome is selected, the subsequent state is $\sum_r \mathcal{I}_r(\rho)$.

§3.4 First Decay and Branching

For the same original-particle preparation, divide the evolution into rounds according to the actual action and storage intervals. When complete records distinguish survival from decay in each round, let $\mathcal{J}_{n,r}$ denote decay branch r in round n, and let $\mathcal{Q}_n$ combine the no-decay outcomes. They cover the complete output on the state space being tracked and satisfy

$$\operatorname{Tr} \mathcal{Q}_n(\sigma) + \sum_r \operatorname{Tr} \mathcal{J}_{n,r}(\sigma) = \operatorname{Tr} \sigma. \tag{3.16}$$

The no-decay branch must close within the space of the tracked parent particle and its participating degrees of freedom. Old auxiliaries and correlations that can affect later actions remain in σ. If no response also includes missed decays, it cannot be identified directly with this physical survival branch; the same record calculation then gives first-detection statistics.

Starting from a normalized initial state ρ_0, continue only the output that has not yet decayed:

$$\sigma_0 = \rho_0, \qquad \sigma_n = \mathcal{Q}_n(\sigma_{n-1}). \tag{3.17}$$

The unnormalized state σ_n describes the uninterrupted-survival history, and its trace gives that history's share. The probability of first entering branch r in round n is

$$\boxed{\begin{aligned} S_n &= \operatorname{Tr} \sigma_n, \\ P(n,r) &= \operatorname{Tr} \mathcal{J}_{n,r}(\sigma_{n-1}). \end{aligned}} \tag{3.18}$$

Inserting σ_{n-1} into the complete balance yields

$$\begin{aligned} \sum_r P(n,r) &= S_{n-1} - S_n, \\ \sum_{n=1}^{M} \sum_r P(n,r) &= 1 - S_M. \end{aligned} \tag{3.19}$$

All first-decay histories and the history that has not yet decayed jointly preserve normalization, as shown in Figure 3-2. The instantaneous parent occupation in Chapter 2 includes possible coherent return. Once a recorded first-decay history has been entered, a subsequently re-formed particle of the same kind continues along that history and is not added back to the original S_n.

If every round has the same action and each new preparation is independent of the joint state of the system, all old environment degrees of freedom, and old records, the same $\mathcal{Q}$ and $\mathcal{J}_r$ can be used on the closed iteration space [1]:

$$\begin{aligned} S_n &= \operatorname{Tr} \mathcal{Q}^n(\rho_0), \\ P(n,r) &= \operatorname{Tr} \mathcal{J}_r \mathcal{Q}^{n-1}(\rho_0). \end{aligned} \tag{3.20}$$

A fixed action can still retain state memory, so the exit share in each round depends on the actual conditional state. If every normalized survival state actually reached has the same no-decay share q and branch shares b_r, then $q + \sum_r b_r = 1$. Iteration gives

$$\boxed{S_n = q^n, \qquad P(n,r) = q^{n-1} b_r.} \tag{3.21}$$

Summing one branch over all first-decay rounds gives its total share B_r. In general,

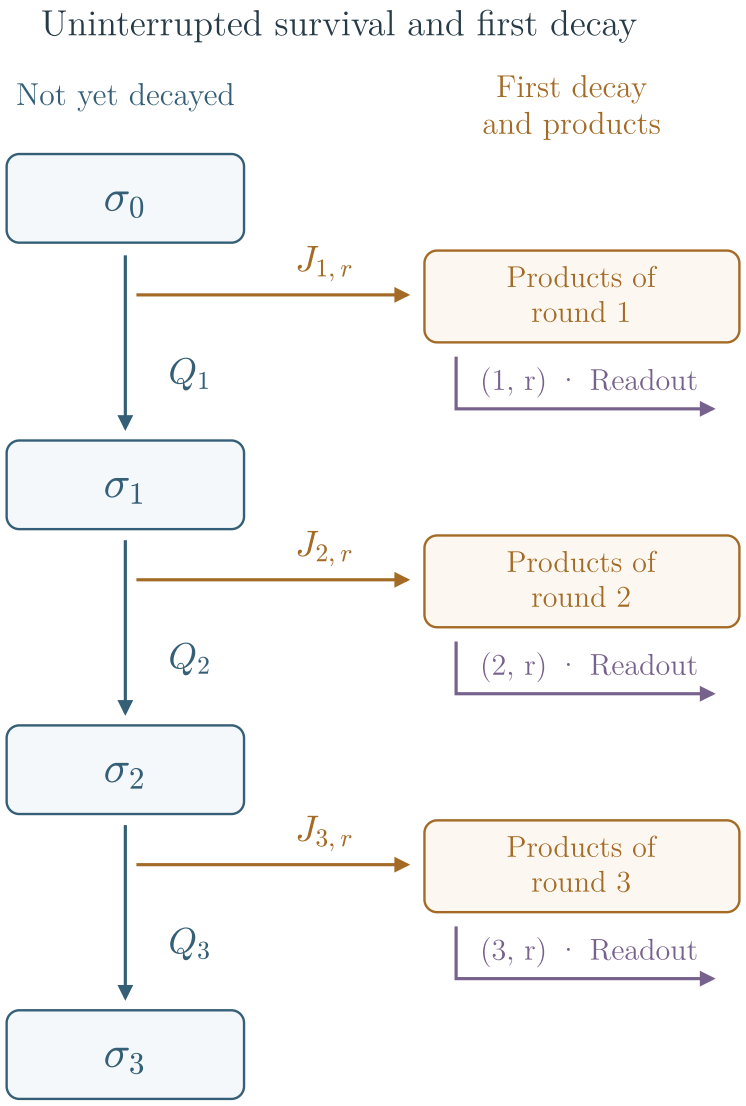


Figure 3-2. Uninterrupted survival and first decay. The left column retains the unnormalized state after each $\mathcal{Q}_n$. The outgoing $\mathcal{J}_{n,r}$ components in round n represent first decay. Records (n, r) label the readout arrows after storage; the products and their subsequent actions remain included. The rows are mutually exclusive first-event histories of the same initial preparation. Branch widths do not assume a constant rate.

$$
\begin{aligned}
B_r &= \sum_{n\geq 1} P(n,r), \\
\sum_r B_r &= 1 - S_\infty .
\end{aligned}
\tag{3.22}
$$

For $0 \leq q < 1$, the geometric sum gives

$$
B_r = \frac{b_r}{1-q}, \qquad \sum_r B_r = 1. \tag{3.23}
$$

If a nonzero long-time survival share remains, B_r is a fraction of all initial preparations. The branching fraction within the decayed sample is obtained by dividing B_r by $1 - S_\infty$.

Actual elapsed durations and time pointers calibrate the physical times of the rounds. The amplitude $B_\alpha(t)$ in Chapter 2 already adds contributions formed at different times coherently. Without the corresponding time record, its integral cannot be replaced by a sum of probabilities over formation times. If an apparatus actually resolves emission intervals, that action must be included in $\mathbb{U}$ and the output recalculated. Distant reception also includes flight and response times, so first reception and emission are calibrated separately.

§3.5 Cascades after Recording

A decay product can itself decay. For example, τ^- first forms $\mu^- \bar{\nu}_\mu \nu_\tau$, and the muon subsequently forms $e^- \bar{\nu}_e \nu_\mu$. The first output determines the muon's motion, spin, and correlations with spectator products; these enter the second decay.

Let $\mathcal{J}_r^{(1)}$ denote a recorded branch of the first decay, $\mathcal{G}$ the complete trace-preserving propagation up to the second interaction, and $\mathcal{J}_s^{(2)}$ a branch of the second decay. On the joint state retaining the required correlations, the unnormalized cascade output is

$$
\boxed{\begin{aligned}
\rho_{rs}^{\mathrm{un}} &= \mathcal{J}_s^{(2)} \circ \mathcal{G} \circ \mathcal{J}_r^{(1)}(\rho_0), \\
P(r,s) &= \operatorname{Tr} \rho_{rs}^{\mathrm{un}}.
\end{aligned}} \tag{3.24}
$$

Writing $P(r) = \operatorname{Tr} \mathcal{J}_r^{(1)}(\rho_0)$ and using the actual conditional output $\rho_r = \mathcal{J}_r^{(1)}(\rho_0)/P(r)$, linearity gives

$$
\begin{aligned}
P(r,s) &= P(r) P(s \mid r), \\
P(s \mid r) &= \operatorname{Tr} \mathcal{J}_s^{(2)} \mathcal{G}(\rho_r).
\end{aligned}
\tag{3.25}
$$

A recorded cascade therefore composes through conditional probabilities. Independent branching fractions can be multiplied only when the relevant correlations no longer affect the second outcome. If the waiting-round counts are also resolved, insert the appropriate no-decay actions before each decay map and compose them in the same order.

If the intermediate process leaves no record distinguishing the paths, amplitudes must still be composed. For two interaction intervals, denote the first amplitude by $a_{mi}^{(1)}$, the intermediate propagation matrix element by $G_{\ell m}$, and the second amplitude by $a_{f\ell}^{(2)}$. When all paths lead to the same remaining output, the total amplitude is

$$\mathcal{A}_{fi} = \sum_{\ell,m} a^{(2)}_{f\ell} G_{\ell m} a^{(1)}_{mi}. \tag{3.26}$$

For partially distinguishable remaining outputs, cross terms are calculated from the output overlaps as in Section 3.2.

First decay, final-state branches, and subsequent cascades thus belong to one history of matter. Accumulating its successive elapsed durations gives the actual waiting time. The relations between survival curves and lifetimes, and between amplitude evolution and spectra, must now be obtained with this time accounting.

§4 Lifetimes and Spectral Lines

The original particle persists before its first decay, and the length of this wait defines a lifetime distribution. The same interaction connecting parent and products also changes the parent's energy response, shifting and broadening its resonance. Accumulating survival histories in actual time and evaluating the energy response of the same generator relate these results. We retain $\hbar$ explicitly in this chapter; energies and elapsed durations use their corresponding physical calibrations.

§4.1 Waiting Time to Decay

The quantity S_n in Chapter 3 is the probability that the same original-particle preparation survives uninterrupted for n rounds. Let $\tau_k > 0$ be the complete actual duration of round k. The endpoints are

$$t_0 = 0, \qquad t_n = \sum_{k=1}^{n} \tau_k. \tag{4.1}$$

Time starts when preparation of the original particle is complete. Evolution, exchange, return, and processing in each round contribute their actual durations [1,2]. Initial loading or final readout is included only in the process in which it occurs. Flight and response times associated with distant detection are calibrated separately and are not added again to the particle's survival time.

Suppose first decay occurs in round N. When the record resolves only the round, initially assign the waiting time to its endpoint, $T = t_N$. A history incurs the wait of round k only if it survives to that round's beginning, so

$$\begin{aligned} T &= \sum_{k\geq 1} \tau_k \mathbf{1}_{\{N\geq k\}}, \\ \Pr(N \geq k) &= S_{k-1}. \end{aligned} \tag{4.2}$$

Taking expectations term by term for these nonnegative contributions gives

$$\boxed{\overline{T} = \mathbb{E}T = \sum_{k\geq 1} \tau_k S_{k-1}.} \tag{4.3}$$

This expression applies even when survival shares differ between rounds. A history that survives forever has $T = \infty$; the accumulated actual duration of an infinite history must also diverge. A nonzero permanent-survival share therefore

makes the mean waiting time over all initial samples infinite. If round durations depend on the preceding record history, the corresponding random durations must remain inside the expectation rather than being replaced directly by unconditional mean values of τ_k.

For a fixed duration τ and the fixed no-decay map $\mathcal{Q}$ of Chapter 3, the mean can also be calculated from the actual conditional states:

$$\begin{aligned} \overline{T} &= \tau \sum_{n\geq 0} \operatorname{Tr} \mathcal{Q}^n(\rho_0) \\ &= \tau \operatorname{Tr}\big[(I-\mathcal{Q})^{-1}(\rho_0)\big] . \end{aligned} \tag{4.4}$$

The second line holds when the series converges on the reachable subspace. It retains the influence of the survival state on later decay without first assuming the same exit share in every round.

§4.2 Survival Curves and Decay Rates

For $S_{n-1} > 0$, the conditional survival and exit shares in round n are

$$q_n = \frac{S_n}{S_{n-1}}, \qquad \epsilon_n = 1 - q_n. \tag{4.5}$$

The ratio ϵ_n/τ_n is the mean decay count rate relative to the sample surviving at the beginning of that round. Relative to all initial samples, it is $(S_{n-1} - S_n)/\tau_n$. For $q_n > 0$, define an exponential rate consistent with both endpoints:

$$\gamma_n = -\frac{\ln q_n}{\tau_n}. \tag{4.6}$$

If the actual conditional action gives a fixed $0 < q < 1$ and every round lasts τ, the geometric law of Chapter 3 and the waiting-time formula become

$$\boxed{\begin{aligned} &S_n = q^n = e^{-\gamma n \tau}, \\ &\overline{T} = \frac{\tau}{1-q}, \qquad \gamma = -\frac{\ln q}{\tau}. \end{aligned}} \tag{4.7}$$

An exponential interpolation $S_{\exp}(t) = e^{-\gamma t}$ connects the endpoints. It passes through all known endpoints, but round-resolved records do not yet specify the within-round decay times. The endpoint-assigned mean waiting time and the exponential time constant are $\tau/(1-q)$ and $1/\gamma$, respectively; the former is strictly larger (Figure 4-1).

Likewise, the first recorded endpoint at which survival is at most one half and the half-life of the exponential interpolation are

$$\begin{aligned} T_{1/2}^{\text{end}} &= \tau \left\lceil \frac{\ln(1/2)}{\ln q} \right\rceil , \\ T_{1/2}^{\exp} &= \frac{\ln 2}{\gamma}. \end{aligned} \tag{4.8}$$

If the per-round exit share $\epsilon = 1 - q$ is small, expanding the logarithm gives

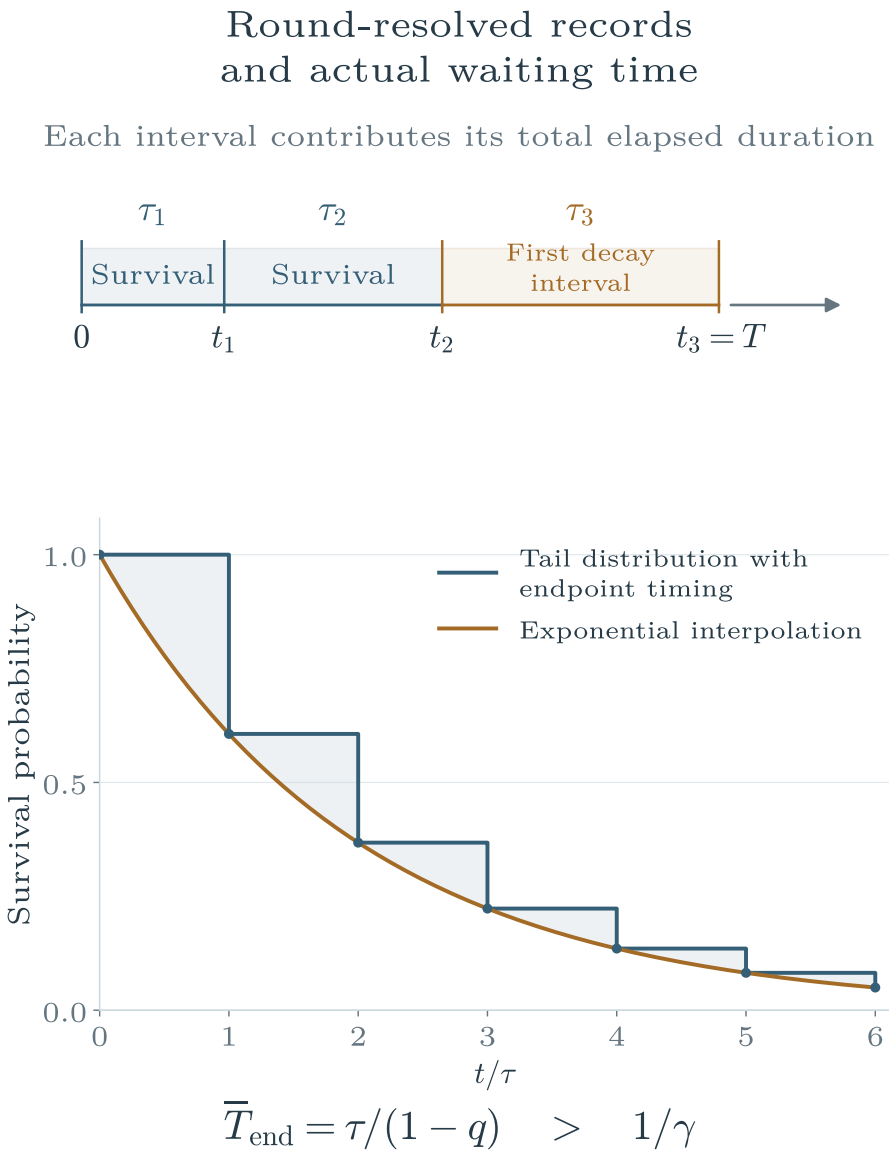


Figure 4-1. Round-resolved records and actual waiting time. The upper segments have widths representing actual durations. First decay is located within the third round, with endpoint assignment $T = t_3$. Below, for fixed q, the blue steps show the endpoint-assigned tail distribution, and the gold curve is an exponential interpolation through all survival endpoints. Their areas over the full time range give $\tau/(1-q)$ and $1/\gamma$. They converge as the resolution interval is shortened at fixed physical decay rate.

$$\begin{aligned}\gamma &= \frac{\epsilon}{\tau}\left(1+\frac{\epsilon}{2}+O(\epsilon^2)\right),\\ \overline{T} &= \frac{\tau}{\epsilon} \simeq \frac{1}{\gamma}.\end{aligned} \tag{4.9}$$

The conditional count rate, exponential rate, and inverse mean lifetime are therefore close in this regime. For $q = 0$, all samples decay in the first round and the endpoint-assigned wait is τ. For $q = 1$, no decay occurs through this process. Changing the actual period also changes the rate calibration: increasing the duration alone produces a $1/\tau$ scaling only if the survival share is held fixed. If conversion continues during the wait, q must be calculated together with it.

§4.3 Transition Amplitudes and Partial Widths

Survival statistics specify how to measure lifetime; the actual rate follows from transfer of the original particle into products. Using the single-parent-state generator of Chapter 2, group orthogonal outgoing final states by decay channel r and define the coupling-weighted energy spectral density

$$\begin{aligned}J_r(E) &= \sum_{\alpha\in r} |g_\alpha|^2 \delta(E-E_\alpha),\\ J(E) &= \sum_r J_r(E).\end{aligned} \tag{4.10}$$

The total amplitude coefficient g_α leading to one complete final state has units of energy, as does J_r. Coherent contributions to that same final state are already included in g_α and must not be added as probabilities over internal intermediate paths.

Restoring $\hbar$, the first-order amplitude of Chapter 2 gives

$$\begin{aligned}P_r^{(1)}(t) = \frac{t^2}{\hbar^2}\int dE\, J_r(E)\\ \times \operatorname{sinc}^2\left[\frac{(E-E_i)t}{2\hbar}\right].\end{aligned} \tag{4.11}$$

A finite interaction time retains a finite energy width. To see how a rate emerges, define

$$\begin{aligned}\delta_t(x) &= \frac{t}{2\pi\hbar}\operatorname{sinc}^2\left(\frac{xt}{2\hbar}\right),\\ \int_{-\infty}^{\infty}\delta_t(x)\,dx &= 1.\end{aligned} \tag{4.12}$$

As the interaction time grows, this normalized kernel concentrates at energy matching. Consider weak coupling, quasi-continuous final states, and J_r smooth over the width $\hbar/t$. For times longer than the final-state correlation time but shorter than the timescales of appreciable depletion and finite-volume return,

$$\begin{aligned}\frac{P_r^{(1)}(t)}{t} &= \frac{2\pi}{\hbar}\int dE\, J_r(E)\delta_t(E-E_i)\\ &\simeq \frac{2\pi}{\hbar}J_r(E_i).\end{aligned} \tag{4.13}$$

This is Fermi's golden rule at the lowest nonzero order. Let γ_r denote the rate per unit time for branch r, and Γ_r its energy partial width. In this approximation,

$$\boxed{\begin{aligned} \gamma_r &= \frac{2\pi}{\hbar} J_r(E_i), \quad \Gamma_r = \hbar\gamma_r, \\ \gamma &= \sum_r \gamma_r, \qquad\quad \Gamma = \sum_r \Gamma_r. \end{aligned}} \tag{4.14}$$

When outgoing final states no longer return appreciably and the retained records correctly distinguish the same decay event, these transfer rates enter the first-event statistics of Chapter 3. Constant branch rates with total rate $\gamma > 0$ give $dS/dt = -\gamma S$, hence $S(t) = e^{-\gamma t}$. Within a resolution interval of duration τ, the survival share and the share first entering channel r are

$$\begin{aligned} q &= e^{-\gamma\tau}, \\ b_r &= \int_0^\tau \gamma_r e^{-\gamma u}\, du \\ &= \frac{\gamma_r}{\gamma}(1 - e^{-\gamma\tau}). \end{aligned} \tag{4.15}$$

Substitution into the geometric branching sum of the preceding chapter gives

$$\boxed{B_r = \frac{b_r}{1-q} = \frac{\gamma_r}{\gamma} = \frac{\Gamma_r}{\Gamma}.} \tag{4.16}$$

Branch proportions are therefore determined by the rates of competing channels. The further approximation $b_r \simeq \gamma_r\tau$ requires $\gamma\tau \ll 1$. Evaluating $J_r(E_i)$ still requires the actual vertex and final-state phase space: internal connections, coupling strength, and product motion jointly determine the lifetime.

§4.4 Resonance Energy and Width

As the parent transfers into products, the products modify its energy response through the same coupling. Take the parent-projected resolvent of the generator in Chapter 2:

$$R_{ii}(z) = \langle i|(z - H)^{-1}|i\rangle. \tag{4.17}$$

Set $(z - H)|x\rangle = |i\rangle$, and expand $|x\rangle$ in the parent state and the eigenbasis of H_F. This yields

$$\begin{aligned} (z - E_i)x_i - \sum_\alpha \overline{g_\alpha} x_\alpha &= 1, \\ (z - E_\alpha)x_\alpha &= g_\alpha x_i. \end{aligned} \tag{4.18}$$

Eliminating the product components gives the parent response

$$\boxed{\begin{aligned} R_{ii}(z) &= \frac{1}{z - E_i - \Sigma(z)}, \\ \Sigma(z) &= \sum_\alpha \frac{|g_\alpha|^2}{z - E_\alpha}. \end{aligned}} \tag{4.19}$$

The self-energy Σ sums contributions entering the products from the parent and connecting back. This block elimination is exact within the finite model, whose self-adjoint generator still has only real eigenvalues. For quasi-continuous final states, $\Sigma(z) = \int J(E')/(z - E')\, dE'$. Its physical boundary value from the upper half-plane decomposes as

$$\begin{aligned} \Sigma(E + i0) &= \Delta(E) - i\pi J(E), \\ \Delta(E) &= \mathrm{PV} \int \frac{J(E')}{E - E'}\, dE'. \end{aligned} \tag{4.20}$$

The real part shifts the resonance, while the imaginary part describes attenuation through open final states. For a weakly coupled process far from thresholds and other resonances, with a self-energy varying slowly across the resonance width, take

$$\begin{aligned} E_R &\simeq E_i + \Delta(E_i), \\ \Gamma &\simeq 2\pi J(E_i) = \hbar\gamma, \\ R_{ii}(E + i0) &\simeq \frac{1}{E - E_R + i\Gamma/2}. \end{aligned} \tag{4.21}$$

In this narrow-resonance approximation, continuing the continuum outgoing response into the lower half-plane gives a resonance pole $z_p \simeq E_R - i\Gamma/2$. Near this peak the parent spectral density is [4]

$$\boxed{\begin{aligned} \mathcal{A}_i(E) &= -\frac{1}{\pi} \operatorname{Im} R_{ii}(E + i0) \\ &\simeq \frac{1}{\pi} \frac{\Gamma/2}{(E - E_R)^2 + (\Gamma/2)^2}. \end{aligned}} \tag{4.22}$$

The peak has full width at half maximum Γ. The corresponding pole contribution in time is

$$\begin{aligned} A_{\mathrm{pole}}(t) &\propto e^{-iE_R t/\hbar} e^{-\Gamma t/(2\hbar)}, \\ |A_{\mathrm{pole}}(t)|^2 &\propto e^{-\Gamma t/\hbar}. \end{aligned} \tag{4.23}$$

Within the same exponential-decay regime, the energy width and probability decay rate obey $\Gamma = \hbar\gamma$, giving the continuous-time mean lifetime $\hbar/\Gamma$. The factor of one half in the amplitude decay exponent follows from taking the squared modulus (Figure 4-2). Very-short-time formation, spectral changes near thresholds, and return in finite systems remain governed by the complete generator. If the self-energy varies appreciably, its energy dependence and the pole residue must be retained; the whole response cannot be extended as a single Lorentzian peak.

Measured final-state spectra also contain channel coupling, preparation, and detection responses. In particular, the continuous electron spectrum in muon decay comes from three-body energy sharing between the electron and two neutrinos. This electron energy distribution differs from the parent particle's resonance line shape. The width of that resonance determines the parent's decay timescale in the regime above; the electron spectrum requires summing over complete three-body final states grouped by electron energy.

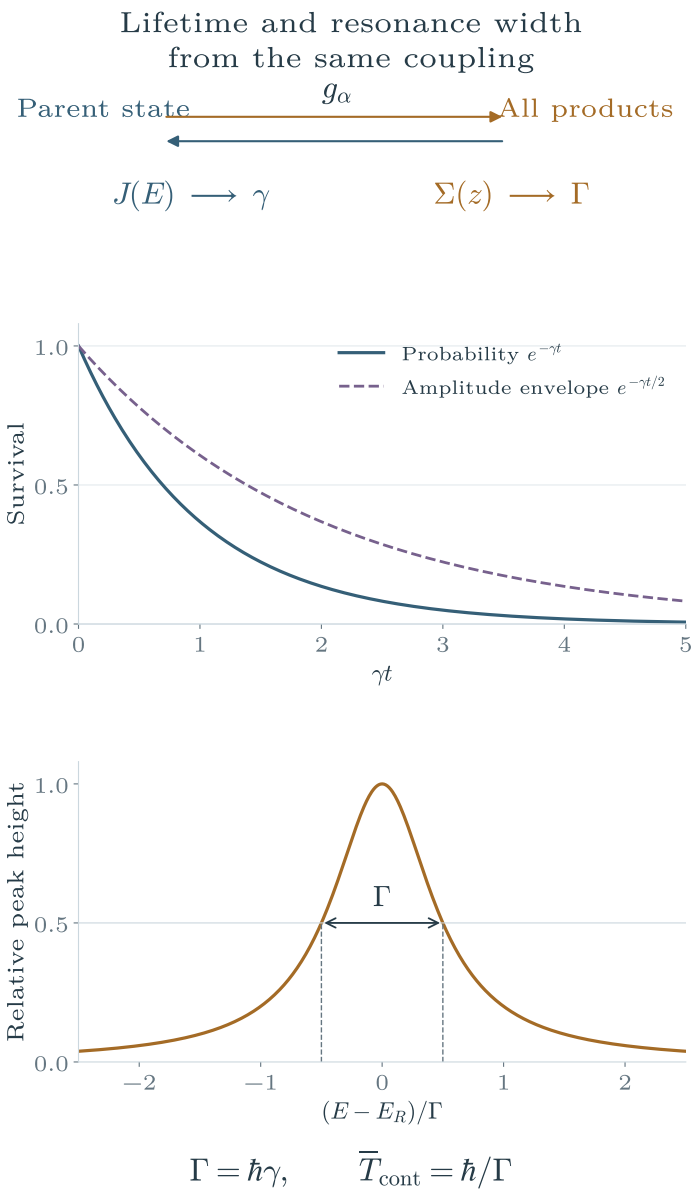


Figure 4-2. Lifetime and resonance width from the same coupling. The coefficients g_α connecting the parent to all products enter both the transfer rate and the self-energy. The middle panel shows probability and amplitude envelopes in time; the lower panel shows the energy response near a narrow resonance, with full width at half maximum $\Gamma = \hbar\gamma$. Dimensionless coordinates illustrate the relation without inserting values for a specific particle.

§4.5 Cycle Phase and Spectral Lines

The same narrow resonance can also be observed cycle by cycle. If the selected survival mode has a common no-decay auxiliary output each round and remains coherent relative to the same phase reference, its unnormalized amplitude can be written as

$$\Lambda = \sqrt{q}\, e^{-iE_R\tau/\hbar}. \tag{4.24}$$

The phase is calibrated over the complete period τ, and E_R is the energy on the corresponding phase branch. After n rounds the survival amplitude is Λ^n, and its probability is q^n. The survival map $\mathcal{Q}$ and its trace in Chapter 3 alone do not supply this complex amplitude by taking a square root of the probability [1,2].

A discrete-time transform of the causal survival sequence gives, by summing over rounds,

$$\begin{aligned} G_\tau(E) &= -\frac{i\tau}{\hbar}\sum_{n\geq 0}\Lambda^n e^{iEn\tau/\hbar} \\ &= \frac{-i\tau/\hbar}{1-\sqrt{q}\, e^{i(E-E_R)\tau/\hbar}}. \end{aligned} \tag{4.25}$$

Using $\gamma = -\ln q/\tau$, the zeros of the denominator give poles at

$$E_k = E_R + \frac{2\pi\hbar k}{\tau} - i\frac{\hbar\gamma}{2}, \qquad k \in \mathbb{Z}. \tag{4.26}$$

The periodic repetition of the real parts comes from discrete-time sampling. Comparison of the same resonance uses the same phase interval. On the real energy axis, the exact intensity response is

$$\begin{aligned} \vartheta &= \frac{(E-E_R)\tau}{\hbar}, \\ |G_\tau(E)|^2 &= \frac{(\tau/\hbar)^2}{(1-\sqrt{q})^2 + 4\sqrt{q}\sin^2(\vartheta/2)}. \end{aligned} \tag{4.27}$$

Near the same peak, with $\gamma\tau \ll 1$ and $|E-E_R|\tau/\hbar \ll 1$, expansion of the denominator gives

$$G_\tau(E) \simeq \frac{1}{E - E_R + i\hbar\gamma/2}. \tag{4.28}$$

The sum over rounds thus recovers the narrow-resonance response of the preceding section. For a general, larger loss per round, the exact real-axis line shape retains its sine term, and its full width at half maximum cannot simply be identified with $\hbar\gamma$.

Coherence can also be affected by actions that leave the particle survival share unchanged. If coherence relative to a nondecaying reference has an additional exponential dephasing rate γ_ϕ, then

$$\begin{aligned} c(t) &\propto e^{-iE_Rt/\hbar}e^{-(\gamma/2+\gamma_\phi)t}, \\ \Gamma_{\mathrm{coh}} &= \hbar\gamma + 2\hbar\gamma_\phi. \end{aligned} \tag{4.29}$$

The second expression is the Lorentzian full width at half maximum of that coherence response. An observed linewidth therefore directly determines particle lifetime only when loss of the measured coherence is controlled by this decay. Survival and coherence must otherwise be calculated separately.

§4.6 Lifetime in Motion

A particle propagates while it survives. Relating rest-frame decay to the waiting time in a frame in which the particle is moving requires decay self-energy and propagation within the same action. Along a uniform propagation direction in the calibrated long-wavelength regime, the real energy has the effective form $E(P) = \sqrt{E_0^2 + c_0^2|P|^2}$, where E_0 is the rest energy and c_0 the corresponding propagation speed scale [2]. If the decay self-energy additionally enters the same complex rest energy in a Lorentz-covariant manner, the narrow-resonance propagation pole is

$$\begin{aligned} z_0 &= E_0 - i\Gamma_0/2, \\ z(P) &= \sqrt{z_0^2 + c_0^2|P|^2}. \end{aligned} \tag{4.30}$$

Choose the propagation branch with positive real part and expand for $\Gamma_0 \ll E_0$:

$$\begin{aligned} z(P) = E(P) - i\frac{\Gamma_0}{2}\frac{E_0}{E(P)} \\ + O(\Gamma_0^2/E_0). \end{aligned} \tag{4.31}$$

The smaller magnitude of the imaginary part slows decay over a fixed laboratory-time interval. Writing the rest decay rate as $\gamma_0 = \Gamma_0/\hbar$, the leading-order result is

$$\boxed{\begin{aligned} \gamma(P) &\simeq \gamma_0 \frac{E_0}{E(P)}, \\ \overline{T}_{\mathrm{cont}}(P) &\simeq \frac{E(P)}{E_0}\overline{T}_{\mathrm{cont}}(0). \end{aligned}} \tag{4.32}$$

The mean lifetime here uses continuous-time statistics in the exponential regime. For a wave packet sufficiently narrow in momentum, group velocity and energy obey

$$\begin{aligned} v_g &= \frac{c_0^2 P}{E(P)}, \\ \gamma_{\mathrm{L}} &= \frac{E(P)}{E_0} = \frac{1}{\sqrt{1 - |v_g|^2/c_0^2}}. \end{aligned} \tag{4.33}$$

The proper time $\theta = t/\gamma_{\mathrm{L}}$ of a uniformly moving particle therefore gives $S_{\mathrm{lab}}(t) \simeq e^{-\gamma_0\theta}$. Within this long-wavelength, covariant narrow-resonance regime, a moving muon has a longer mean waiting time in the laboratory and, correspondingly, a longer mean flight distance [3].

Lifetime, branching, and resonance width have thus been related to the same actual transfer coefficients and their final-state distribution. For muon decay, the internal connection must next be realized as a complete leptonic vertex, with the electron and both neutrinos included in the phase-space integral. The same calculation determines the total lifetime and the electron energy spectrum.

§5 Decay Phenomena

Muon decay produces a continuous electron spectrum, charged pions predominantly form a muon and a neutrino, and an alpha cluster inside a nucleus becomes a free product beyond the barrier. These processes all reduce the uninterrupted-survival share of the original matter. Their lifetimes differ because of their transfer actions, final states, and propagation. We now realize the emission coefficients of Chapter 2 through specific effective vertices and calculate their consequences using the rate relations of Chapter 4.

In the long-wavelength regime considered, adopt relativistically normalized outgoing multiparticle states. We set $\hbar = c = 1$; Γ always denotes an energy width, and conversion to seconds restores $\overline{T} = \hbar/\Gamma$. Momentum states are normalized by [3]

$$\begin{aligned} &\langle \mathbf{p}, s | \mathbf{p}', s' \rangle \\ &= 2E(2\pi)^3 \delta^3(\mathbf{p} - \mathbf{p}') \delta_{ss'}. \end{aligned} \tag{5.1}$$

This convention must be related to the normalized finite modes of Chapter 2. Let $\mathcal{V}$ be the periodic-box volume, with one fixed parent mode and n distinguishable outgoing modes. In the relativistic effective description, take a translation-invariant, Lorentz-covariant vertex. Each external leg contributes $(2E\mathcal{V})^{-1/2}$; the spatial integral supplies the volume and a momentum Kronecker delta. The finite-mode coefficient is related to the invariant amplitude by

$$\begin{aligned} g_\alpha = &\frac{\mathcal{M}_\alpha}{\mathcal{V}^{(n-1)/2} \sqrt{2E_i \prod_a 2E_a}} \\ &\times \delta_{\mathbf{P}, \sum_a \mathbf{p}_a}. \end{aligned} \tag{5.2}$$

A common phase has been absorbed into $\mathcal{M}_\alpha$. The coefficient g_α has units of energy, while $\mathcal{M}_\alpha$ has units of energy to the power $3 - n$ in this convention. External-state normalization connects them. Insert this relation into Fermi's golden rule from Chapter 4, where $\langle \cdot \rangle_{\text{in}}$ averages only over the initial preparation and the sum over α already includes final spins. Replacing the dense box momentum modes by continuum integrals gives

$$\begin{aligned} \Gamma_r = &2\pi \sum_{\alpha \in r} \langle |g_\alpha|^2 \rangle_{\text{in}} \\ &\times \delta\left(E_i - \sum_a E_a \right), \\ \sum_{\mathbf{p}} &\longrightarrow \mathcal{V} \int \frac{d^3p}{(2\pi)^3}, \\ &\delta_{\mathbf{P}, \sum_a \mathbf{p}_a} \\ \longrightarrow &\frac{(2\pi)^3}{\mathcal{V}} \delta^3\left(\mathbf{P} - \sum_a \mathbf{p}_a \right). \end{aligned} \tag{5.3}$$

The Kronecker delta equals its own square. Volume factors from the momentum sums cancel those in g_α and the momentum constraint. Combining energy and

momentum constraints into a four-dimensional delta function, with $E_i = M$ in the parent's rest frame, yields the usual decay phase-space formula:

$$\begin{aligned} d\Phi_n &= (2\pi)^4 \delta^4\left(P - \sum_a p_a\right) \\ &\quad \times \prod_a \frac{d^3 p_a}{(2\pi)^3\, 2E_a}, \\ \Gamma_r &= \frac{1}{2M} \int d\Phi_r\, \overline{|\mathcal{M}_r|^2}. \end{aligned} \tag{5.4}$$

The bar denotes an average over initial spins according to the actual preparation and a sum over final spins. Identical indistinguishable particles in the final state require the appropriate symmetry factor for the counting convention. The product-formation amplitudes of Chapter 2 and the specific spectra, branches, and lifetimes thus use the same transfer relation. Chiral vertex forms, strength matching, and composite-matter responses are specified where first used.

§5.1 Muon Lifetime and Electron Energy Spectrum

The three products of muon decay are $\mu^- \to e^- + \bar{\nu}_e + \nu_\mu$. Chapter 2 constructed an emission term that annihilates the parent and creates all three products simultaneously. We now take its spin and motion coefficient to have the low-energy charged-current $V - A$ contact form [5]:

$$\begin{aligned} \mathcal{M}_\mu = \frac{G_F}{\sqrt{2}} &[\bar{u}_{\nu_\mu} \gamma^a (1 - \gamma^5) u_\mu] \\ &\times [\bar{u}_e \gamma_a (1 - \gamma^5) v_{\bar{\nu}_e}]. \end{aligned} \tag{5.5}$$

This contact vertex specifies the chiral structure adopted here. Its strength is normalized to the electroweak energy scale v of [2] by

$$\boxed{\begin{gathered} G_F = \frac{1}{\sqrt{2}\, v^2}, \\ v = 246.191373933\ldots \text{ GeV}. \end{gathered}} \tag{5.6}$$

Here v is an energy scale; actual cycle durations are still accounted for as in Chapter 4. This matching gives $G_F = 1.16664665 \times 10^{-5}$ GeV^{-2}. The following calculation is at tree level for this contact interaction. Neutrino-mass effects on the total width are neglected, while the electron mass is retained.

Let P, p, k, k' be the four-momenta of the parent, electron, muon neutrino, and electron antineutrino, respectively, and set $M = m_\mu$, $m = m_e$. Spinor completeness and Dirac traces give the initial-spin-averaged result

$$\overline{|\mathcal{M}_\mu|^2} = 64 G_F^2 (P \cdot k')(p \cdot k). \tag{5.7}$$

With $q = p + k$ and $s = q^2$, we have $2P \cdot k' = M^2 - s$ and $2p \cdot k = s - m^2$. Factoring the three-body phase space through this intermediate invariant gives

$$
\begin{aligned}
d\Phi_3 &= \frac{ds}{2\pi} d\Phi_2(P; q, k') \\
&\quad \times d\Phi_2(q; p, k), \\
\int d\Phi_2(P; q, k') &= \frac{M^2 - s}{8\pi M^2}, \\
\int d\Phi_2(q; p, k) &= \frac{s - m^2}{8\pi s}.
\end{aligned}
\tag{5.8}
$$

The spin-averaged squared amplitude depends only on s, so the angular integrals can be completed directly:

$$
\begin{aligned}
\Gamma_\mu &= \frac{G_F^2}{16\pi^3 M^3} \\
&\quad \times \int_{m^2}^{M^2} \frac{(M^2 - s)^2 (s - m^2)^2}{s}\, ds.
\end{aligned}
\tag{5.9}
$$

Writing $r = m^2/M^2$ and $u = s/M^2$, the dimensionless integral is

$$
\begin{aligned}
f(r) &= 12 \int_r^1 \frac{(1-u)^2 (u-r)^2}{u}\, du \\
&= 1 - 8r + 8r^3 - r^4 - 12r^2 \ln r.
\end{aligned}
\tag{5.10}
$$

The lifetime and finite-electron-mass correction thus follow from the same phase-space integral:

$$
\boxed{\Gamma_\mu = \frac{G_F^2 m_\mu^5}{192\pi^3} f\left(\frac{m_e^2}{m_\mu^2}\right).}
\tag{5.11}
$$

Using $m_\mu = 105.6553711\ldots$ MeV and $m_e = 0.5110122512\ldots$ MeV from [2] gives the tree-level mean lifetime 2.187063×10^{-6} s. The PDG 2025 value is $(2.1969811 \pm 0.0000022) \times 10^{-6}$ s [6], a central-value difference of approximately -0.45%. Electromagnetic radiation and virtual corrections have not yet been included; the difference must be assessed together with those corrections and the input parameters.

The same matrix element determines the electron spectrum. First take $m = 0$, write $x = 2E_e/M$, $y = 2E_{\bar{\nu}_e}/M$, and define $\Gamma_* = G_F^2 M^5/(192\pi^3)$. The three-body energy region is $0 \le x \le 1$, $1 - x \le y \le 1$, with differential width

$$
\begin{aligned}
\frac{d^2\Gamma}{dx\, dy} &= 12\Gamma_* y(1-y), \\
\frac{d\Gamma}{dx} &= 12\Gamma_* \int_{1-x}^1 y(1-y)\, dy \\
&= 2\Gamma_* x^2 (3 - 2x).
\end{aligned}
\tag{5.12}
$$

The electron receives only part of the available energy; the two neutrinos share the rest. Each electron energy record gives one reading, while repeated samples form a continuous spectrum. Retaining the electron mass, the same integration yields

$$\begin{aligned}\frac{d\Gamma}{dx} &= 2\Gamma_*\sqrt{x^2-4r} \\ &\quad\times[x(3-2x)+r(3x-4)], \\ 2\sqrt{r} &\le x \le 1+r.\end{aligned} \tag{5.13}$$

Integration over x again gives $\Gamma_* f(r)$, checking the total lifetime and spectrum together (Figure 5-1).

For a parent μ^- with polarization $\mathcal{P}_\mu$, let the unit spin four-vector satisfy $P\cdot s = 0$, $s^2 = -1$. Replace the unpolarized average by the spin density matrix $(\not{P}+M)(1+\mathcal{P}_\mu\gamma^5\not{s})/2$. Neglecting the electron mass gives [5]

$$\begin{aligned}\frac{d^2\Gamma}{dx\,d\cos\theta} &= \Gamma_* x^2[3-2x \\ &\quad+\mathcal{P}_\mu\cos\theta(1-2x)].\end{aligned} \tag{5.14}$$

Here θ is the angle between electron momentum and parent spin. Spatial inversion reverses momentum but preserves spin as an axial vector, so the angular correlation changes sign. The adopted left-chiral vertex is not paired with an equal-strength right-chiral term, and the distribution therefore violates parity. High-energy electrons are preferentially emitted opposite to the μ^- spin. Muon polarization prepared in pion decay can thus pass through the conditional output of Chapter 3 into the subsequent electron angular distribution.

§5.2 Tau Decay and Competing Branches

The tau has more available energy, so the same charged current forms both leptonic and hadronic final states. Replacing the parent and final-lepton masses in the muon calculation gives the two leptonic partial widths:

$$\begin{aligned}\Gamma_{\tau\ell} &= \Gamma_{\tau*} f_\ell, \qquad \ell = e,\mu, \\ \Gamma_{\tau*} &= \frac{G_F^2 m_\tau^5}{192\pi^3}, \qquad f_\ell = f(m_\ell^2/m_\tau^2).\end{aligned} \tag{5.15}$$

The hadronic branch requires a sum of quark-current matrix elements over actual hadronic final states. Writing the leptonic current as L_a and the hadronic charged current as J_h^a,

$$\begin{aligned}\mathcal{M}_X &= \frac{G_F}{\sqrt{2}} L_a \langle X|J_h^a|0\rangle, \\ W^{ab}(q) &= \sum_X \int d\Phi_X(q) \\ &\quad\times\langle 0|J_h^{a\dagger}|X\rangle\langle X|J_h^b|0\rangle.\end{aligned} \tag{5.16}$$

Here X is a color-singlet hadronic output; its energy, thresholds, and internal interactions enter W^{ab}. Contracting this response with the leptonic tensor and integrating the remaining phase space gives $\Gamma_{\tau\mathrm{had}}$. Define its inclusive weight relative to $\Gamma_{\tau*}$ as w_{had}. The total width and branches become

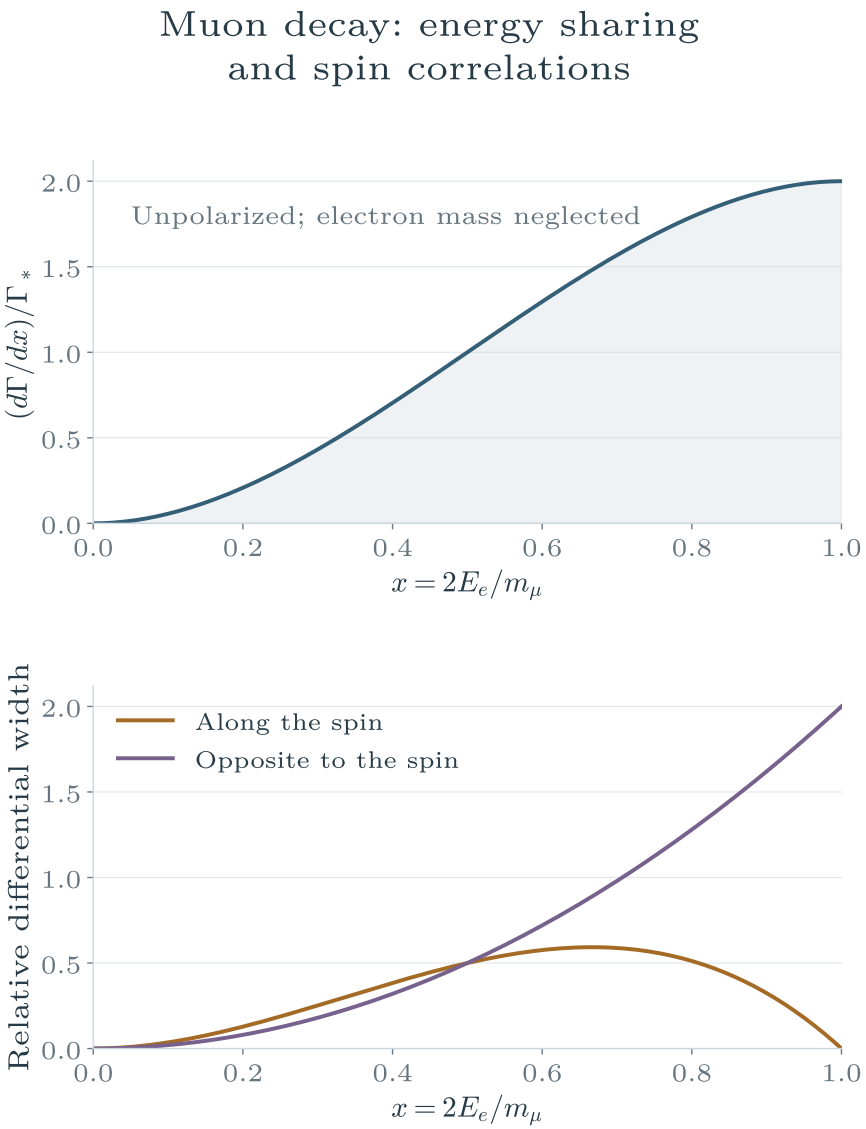


Figure 5-1. Energy sharing and angular distributions in muon decay. The upper panel shows the normalized continuous electron spectrum with electron mass neglected. The lower panel uses the same chiral vertex to compare differential spectra along and opposite to the spin of a fully polarized μ^-. The angular distribution follows from the transition amplitude and changes with the spin preparation.

$$\boxed{\begin{aligned}\Gamma_\tau &= \Gamma_{\tau*}(f_e + f_\mu + w_{\text{had}}),\\ B_{\tau\ell} &= \frac{f_\ell}{f_e + f_\mu + w_{\text{had}}},\\ B_{\tau\text{had}} &= \frac{w_{\text{had}}}{f_e + f_\mu + w_{\text{had}}}.\end{aligned}} \tag{5.17}$$

To give the inclusive response a calculable realization, adopt the following order-by-order spectral-weight model. Two participating lines and a conjugate pair of insertion members provide four equally weighted configurations, two with nonzero channel response. Color contraction gives three shares and electromagnetism one. Taking the basic couplings of [2] as inputs, choose the coefficient per order as

$$\begin{aligned}\xi &= \frac{3}{2}\alpha_s^{(0)} + \frac{1}{2}\alpha_{\text{EM}}^{(0)},\\ \alpha_s^{(0)} &= 0.1149199747\ldots,\\ \alpha_{\text{EM}}^{(0)} &= 0.00792698931\ldots.\end{aligned} \tag{5.18}$$

Further specify mutually orthogonal response outputs at different orders, with the same baseline spectral shape $J_0(E)$ on the same energy shell and weights $J_{\text{had}}^{(n)}(E) = 3\xi^n J_0(E)$. Normalize the baseline to one massless leptonic channel, $2\pi J_0(m_\tau) = \Gamma_{\tau*}$. Summing orthogonal final states as in Chapter 4 gives

$$\begin{aligned}J_{\text{had}}(E) &= 3J_0(E)\sum_{n\geq 0}\xi^n,\\ w_{\text{had}} &= \frac{3}{1-\xi} = 3.642294867\ldots.\end{aligned} \tag{5.19}$$

Here ξ is a specified relative spectral weight. If it were instead a probability of continued residence, terminating histories would also carry $1-\xi$, and their sum would be one; that would not imply a width enhancement. Matching the full hadronic interaction to this spectral-weight model remains to be calculated. Its consequences can first be compared with actual branches.

Taking $m_\tau = 1776.926974\ldots$ MeV gives Table 5-1. The experimental partial widths are converted from PDG 2025 branching fits and mean lifetime and are not independent measurements [7].

Quantity	Tree-level / specified-response result	PDG 2025 comparison
$\Gamma_{\tau e}$ (GeV)	4.05019×10^{-13}	4.04722×10^{-13}
$\Gamma_{\tau\mu}$ (GeV)	3.93907×10^{-13}	3.93839×10^{-13}
$B_{\tau\mu}$	0.173212	0.1737 ± 0.0004
$\overline{T}_\tau$ (s)	2.89435×10^{-13}	$(2.903 \pm 0.005) \times 10^{-13}$

Table 5-1. Tau partial widths, branching fractions, and mean lifetime. The theoretical column uses the tree-level vertex and hadronic response specified here; the comparison column uses the same PDG edition throughout.

The tau decays faster than the muon both because its leptonic partial widths contain the fifth power of the parent mass and because additional hadronic outputs

draw on the survival share. The leptonic ratio $\Gamma_{\tau\mu}/\Gamma_{\tau e} = f_\mu/f_e$ cancels the inclusive hadronic response, while total lifetime and absolute branching fractions test that response. Hadronic thresholds and spectral shapes are determined by the actual W^{ab}; radiative corrections must be included separately at the corresponding vertices and in the outgoing states.

§5.3 Beta Decay, Endpoints, and Nucleon Response

A free neutron decays through $n \to p + e^- + \bar{\nu}_e$. The weak vertex acts within the nucleon, and both its vector and axial-vector responses control the amplitudes of the outgoing electron and neutrino. In the allowed-transition approximation at small momentum transfer, write [8]

$$\begin{aligned}\mathcal{M}_\beta = \frac{G_F V_{ud}}{\sqrt{2}}[\bar{u}_p(g_V\gamma^a - g_A\gamma^a\gamma^5)u_n] \\ \times [\bar{u}_e\gamma_a(1-\gamma^5)v_\nu].\end{aligned} \tag{5.20}$$

Use $g_V = 1$ and $g_A > 0$ as the normalization here, with g_A the axial-vector to vector response ratio. For large nucleon mass, the leading components are $J^0_V \propto g_V$ and $\mathbf{J}_A \propto g_A\boldsymbol{\sigma}$. Averaging spins and integrating unobserved directions removes the electron–neutrino angular correlation term, while the three spatial components give the weight $g_V^2 + 3g_A^2$. Neglecting recoil and electromagnetic interactions, the three-body integral reduces to

$$\begin{aligned}\frac{d\Gamma_n}{dE_e} = \frac{G_F^2|V_{ud}|^2}{2\pi^3}(g_V^2 + 3g_A^2) \\ \times p_e E_e (E_0 - E_e)^2, \\ E_0 = m_n - m_p.\end{aligned} \tag{5.21}$$

The nucleon carries recoil momentum; at leading order its recoil energy is small compared with the energy shared by the electron and neutrino. The electron's total energy varies continuously from m_e to E_0, with kinetic endpoint $Q_\beta = E_0 - m_e$. Finite masses and recoil shift the exact endpoint, so the allowed approximation does not replace a precision endpoint fit.

For allowed nuclear transitions, let C_β include the relevant nuclear matrix element and $F(Z, E_e)$ the correction from the final-state Coulomb field acting on the electron. Different neutrino mass states are orthogonal and carry production weights $|U_{ej}|^2$. The electron spectrum is

$$\begin{aligned}\frac{d\Gamma}{dE_e} = C_\beta F(Z, E_e) p_e E_e \\ \times \sum_j |U_{ej}|^2 \, \omega\sqrt{\omega^2 - m_{\nu_j}^2} \\ \times \Theta(\omega - m_{\nu_j}), \\ \omega = E_0 - E_e.\end{aligned} \tag{5.22}$$

This gives the continuous spectrum, its endpoint, and the threshold of each mass state. In the massless limit, removing the electron density of states and Coulomb response gives the Kurie quantity $K(E_e) = \sqrt{(d\Gamma/dE_e)/(F p_e E_e)}$, proportional to

$E_0 - E_e$. Nonzero neutrino mass bends it near the endpoint. Figure 5-2 places energy sharing alongside this endpoint structure.

Different allowed nuclear transitions can also be used to compare the same vector interaction. For superallowed $0^+ \to 0^+$ transitions the axial-vector term does not contribute. Define the Fermi matrix element as $M_F = \langle f| \sum_a \tau_a^{\pm} |i\rangle$. At the order neglecting recoil, neutrino masses, and radiative corrections, define the dimensionless phase-space integral

$$\begin{aligned} f_\beta &= \frac{1}{m_e^5} \int_{m_e}^{E_0} F(Z, E_e) p_e E_e \\ &\quad \times (E_0 - E_e)^2 \, dE_e, \\ \Gamma_F &= \frac{G_F^2 |V_{ud}|^2 g_V^2 |M_F|^2 m_e^5}{2\pi^3} \, f_\beta. \end{aligned} \tag{5.23}$$

The partial half-life of this branch is $t_{1/2}^{(F)} = \hbar \ln 2 / \Gamma_F$, so

$$\boxed{f_\beta t_{1/2}^{(F)} = \frac{2\pi^3 \hbar \ln 2}{G_F^2 |V_{ud}|^2 g_V^2 |M_F|^2 m_e^5}.} \tag{5.24}$$

In the isospin-symmetric limit, $T_{\text{iso}} = 1$ isobaric analogue states have $|M_F|^2 = 2$. Available energies differ among nuclei, making their partial half-lives vary with phase space. Multiplying the partial half-life by f_β permits comparison of the common vector normalization. Precision comparisons use $\mathcal{F}t$, including radiative and isospin-breaking corrections [19].

Neutron decay also contains an axial-vector response, which can be connected further to the internal spectral data of [2]. Adopt a normalized two-generation link for the first and second charged-current generations, allowing its formation link λ to enter the weak vertex:

$$\begin{aligned} |V_{us}| &= \lambda = 0.2247147851\ldots, \\ |V_{ud}| &= \sqrt{1 - \lambda^2}. \end{aligned} \tag{5.25}$$

This is a correspondence between the formation link and the charged current; the link alone does not determine a full three-generation mixing matrix. For the axial-vector response, adopt an in-phase return model with color insertions. Each return multiplies by $a = 3\alpha_s^{(0)}/2$, the total return satisfies $R_A = 1 + aR_A$, and one color-singlet endpoint response $\alpha_s^{(0)}/\sqrt{3}$ is added. Thus

$$\boxed{\begin{aligned} g_A &= \frac{1}{1 - 3\alpha_s^{(0)}/2} + \frac{\alpha_s^{(0)}}{\sqrt{3}} \\ &= 1.274633018\ldots, \\ |V_{ud}| &= 0.974424582\ldots. \end{aligned}} \tag{5.26}$$

The return coefficient, phase, and single endpoint term specify the normalization of this nucleon-response model. PDG uses a signed axial-vector to vector ratio. In the vertex convention of this section, the appropriate comparison is its magnitude, $|g_A/g_V| = 1.2753 \pm 0.0013$ [16]. The corresponding comparison value for $|V_{ud}|$ is 0.97367 ± 0.00032 [17]; the difference from the model central value exceeds the quoted experimental uncertainty.

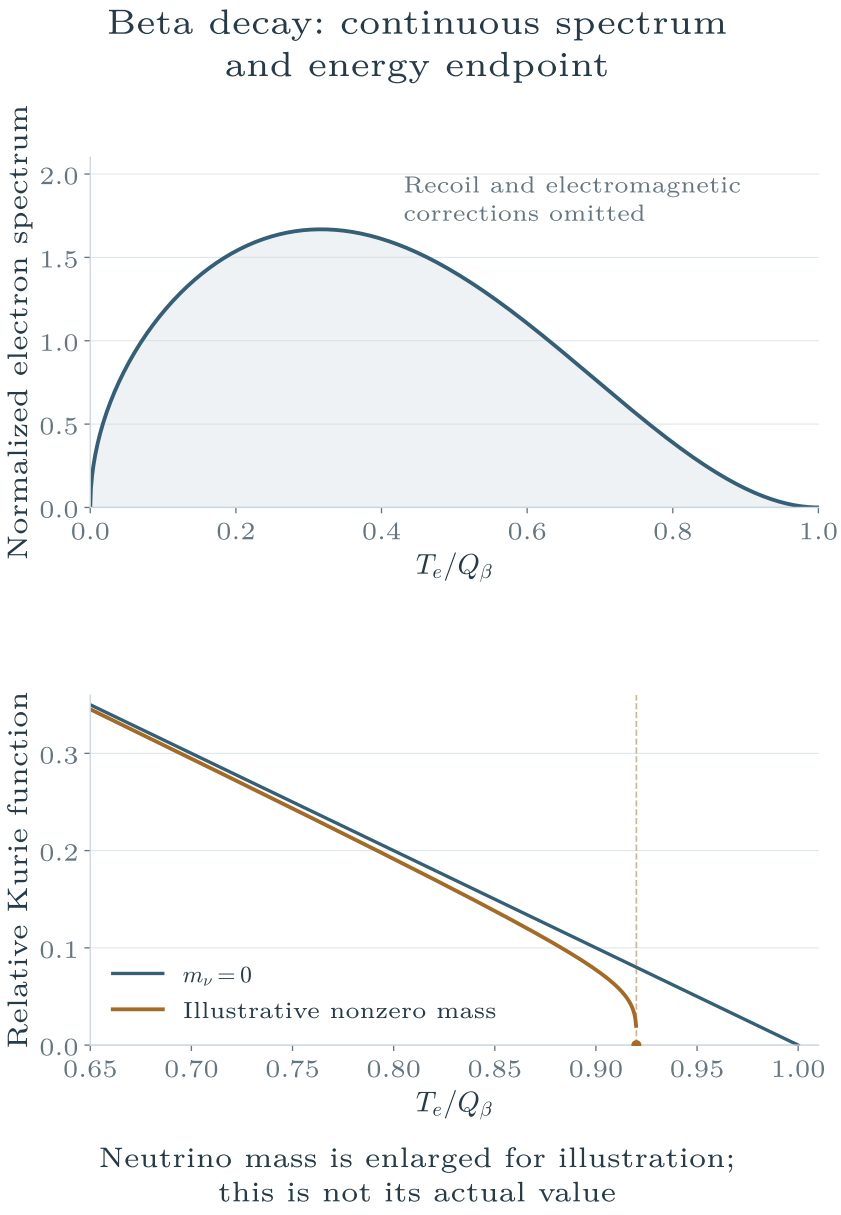


Figure 5-2. Continuous beta spectrum and endpoint. The upper panel shows the allowed neutron spectrum without recoil or electromagnetic corrections; the electron and neutrino share the available energy. Below, one nonzero neutrino mass is enlarged for illustration: the spectrum opens only for $E_0 - E_e \geq m_\nu$, and the Kurie curve departs from a straight line near the endpoint. The illustrated mass is not an actual neutrino-mass value.

Writing electron energy as $w = E_e/m_e$ and $w_0 = E_0/m_e$, the total width is

$$\begin{aligned} f_0(w_0) &= \int_1^{w_0} w\sqrt{w^2-1}(w_0-w)^2\,dw, \\ \Gamma_n &= \frac{G_F^2|V_{ud}|^2 m_e^5}{2\pi^3} \\ &\quad \times (1+3g_A^2) f_0(w_0). \end{aligned} \tag{5.27}$$

With $E_0 = 1.303630499\ldots$ MeV from [2], this gives $f_0 = 1.72268806\ldots$, $Q_\beta = 0.79261825\ldots$ MeV, and $\overline{T}_n = 895.71$ s in this approximation. The result lies on the scale of minutes. Its difference from the PDG 2025 value 878.4 ± 0.5 s [16] involves the present mass difference, nucleon response, recoil, and electromagnetic terms; a precision comparison must account for them.

Polarized-nucleon angular distributions retain interference between vector and axial-vector contributions. With the positive-g_A convention above, the leading neutron electron asymmetry is

$$\begin{aligned} d\Gamma &\propto d\Gamma_0 \left(1 + A_n \mathcal{P}_n \frac{p_e}{E_e}\cos\theta\right), \\ A_n &= -\frac{2g_A(g_A-1)}{1+3g_A^2} = -0.119187\ldots. \end{aligned} \tag{5.28}$$

Electron counts differ along and opposite to the neutron spin. Angular asymmetry and total lifetime test the interference term and sum of squares of the same nucleon response, respectively. The parity asymmetry follows from the chiral structure of the contact vertex and its correlation with neutron spin.

For some nuclei, the single-beta final state lies too high in energy while the double-beta final state is open. Two weak actions can then connect the states coherently. For two-neutrino double-beta decay, the second-order amplitude retains denominators containing intermediate nuclear states and lepton energies:

$$\begin{aligned} \mathcal{T}_{fi}^{(2)} &= \sum_m \frac{\langle f|V_2|m\rangle\langle m|V_1|i\rangle}{E_i - E_m + i0} \\ &\quad + \text{other time orderings}. \end{aligned} \tag{5.29}$$

The complete intermediate state m includes leptons already formed in that time ordering. After $|\mathcal{T}_{fi}^{(2)}|^2$ is integrated over the four-lepton phase space, the resulting rate shows how the two contact coefficients, nuclear energy gaps, and restricted phase space jointly produce a very long lifetime. Such two-neutrino decay has been observed by direct counting [18]. If the intermediate nucleus actually forms and leaves a record, use the cascade conditional outputs and successive waiting times of Chapter 3 instead. Neutrinoless double-beta decay requires an actual lepton-number-violating action and does not follow automatically from two charged-current connections in this section.

§5.4 Monoenergetic Pion Products and Helicity Suppression

A charged pion is composite matter with spin zero. The final state of $\pi^- \to \ell^- \bar{\nu}_\ell$ contains two particles. In its rest frame, energy-momentum conservation fixes their momentum magnitude and the lepton energy:

$$p_\ell = \frac{m_\pi^2 - m_\ell^2}{2m_\pi}, \qquad E_\ell = \frac{m_\pi^2 + m_\ell^2}{2m_\pi}. \tag{5.30}$$

Unlike the three-body continuous spectrum of muon decay, this lepton has a definite energy in the narrow-parent-state approximation. The pion axial-current matrix element defines the decay constant f_π [9]:

$$\langle 0|\bar{u}\gamma^a\gamma^5 d|\pi^-(P)\rangle = i f_\pi P^a. \tag{5.31}$$

Insert it into the same charged current, set $P = p + k$, and use the outgoing Dirac equation and massless neutrino equation:

$$\begin{aligned} \mathcal{M}_{\pi\ell} &= \frac{iG_F V_{ud} f_\pi}{\sqrt{2}} \bar{u}_\ell \, \not{P}(1-\gamma^5) v_\nu \\ &= \frac{iG_F V_{ud} f_\pi m_\ell}{\sqrt{2}} \bar{u}_\ell (1-\gamma^5) v_\nu. \end{aligned} \tag{5.32}$$

The momentum contraction becomes a lepton-mass factor, so the amplitude vanishes as $m_\ell \to 0$. The spin-zero parent and the selected chiral final state cannot retain a nonzero amplitude in the massless limit. This is helicity suppression. Summing spins gives

$$\begin{aligned} \sum |\mathcal{M}_{\pi\ell}|^2 = 2G_F^2 |V_{ud}|^2 f_\pi^2 \\ \times m_\ell^2 (m_\pi^2 - m_\ell^2). \end{aligned} \tag{5.33}$$

Substitution into the two-body width $p_\ell \sum |\mathcal{M}|^2/(8\pi m_\pi^2)$ yields

$$\boxed{\Gamma_{\pi\ell} = \frac{G_F^2 |V_{ud}|^2 f_\pi^2}{8\pi} m_\pi m_\ell^2 \left(1 - \frac{m_\ell^2}{m_\pi^2}\right)^2.} \tag{5.34}$$

Although the lighter electron has more available phase space, the mass factor in its amplitude is smaller, so the muon branch dominates. Taking the ratio cancels the common hadronic matrix element:

$$R_{e/\mu}^{(0)} = \frac{m_e^2}{m_\mu^2} \frac{(1 - m_e^2/m_\pi^2)^2}{(1 - m_\mu^2/m_\pi^2)^2}. \tag{5.35}$$

Using the common pion mass 138.2894763… MeV of [2], which does not yet separate the electromagnetic mass splitting, gives $R_{e/\mu}^{(0)} = 1.349887 \times 10^{-4}$. This is a tree-level ratio with the common mass input. A more precise comparison requires the charged-pion mass and radiative corrections separately. The absolute pion lifetime additionally requires an independently determined f_π.

§5.5 Strange Particles and Neutral-Meson Mixing

Strong interactions can produce a pair of particles containing strange quarks while preserving net flavor. Some lighter strange hadrons have no open strong-decay channel and change their quark composition through a weak vertex. Strong production and subsequent weak decay involve different connections, so rapid production can leave a long-lived particle.

In the two-generation charged-current correspondence adopted in Section 5.3, a strangeness-changing amplitude contains V_{us}. For identical kinematics and hadronic responses, the width ratio contains

$$\frac{|V_{us}|^2}{|V_{ud}|^2} = \frac{\lambda^2}{1-\lambda^2} = 0.05318\ldots. \tag{5.36}$$

Actual kaon and pion processes have different masses, form factors, and phase spaces, which must also be included. This ratio isolates only the suppression due to the link.

Neutral mesons can also undergo particle–antiparticle mixing before decay. Extending the single parent of Chapter 4 to $\{|K^0\rangle, |\bar{K}^0\rangle\}$, their common final states generate a matrix self-energy:

$$\begin{aligned} \Sigma_{ab}(z) &= \sum_{\alpha} \frac{\overline{g_{\alpha a}} g_{\alpha b}}{z - E_\alpha}, \\ H_{\text{eff}} &= M - \frac{i}{2}\mathbf{\Gamma}, \\ M = M^\dagger, &\quad \mathbf{\Gamma} = \mathbf{\Gamma}^\dagger. \end{aligned} \tag{5.37}$$

Here M includes the principal-value real part of the self-energy, and the width matrix is

$$\Gamma_{ab} = 2\pi \sum_{\alpha} \overline{g_{\alpha a}} g_{\alpha b} \delta(E - E_\alpha). \tag{5.38}$$

This matrix is positive semidefinite; its off-diagonal entries retain coherent sums over common final states. With a narrow-resonance effective action and equal diagonal entries, the propagation eigenstates can be written as $|K_{S,L}\rangle = p_K|K^0\rangle \pm q_K|\bar{K}^0\rangle$, with $|p_K|^2 + |q_K|^2 = 1$ and eigenvalues $z_{S,L} = m_{S,L} - i\Gamma_{S,L}/2$. Starting from a pure K^0 [10],

$$\begin{aligned} |K^0(t)\rangle &= g_+(t)|K^0\rangle + \frac{q_K}{p_K} g_-(t)|\bar{K}^0\rangle, \\ g_\pm(t) &= \frac{e^{-iz_S t} \pm e^{-iz_L t}}{2}. \end{aligned} \tag{5.39}$$

The mixing probability follows from coherent subtraction of the two propagation amplitudes:

$$\begin{aligned} P_{K^0\to\bar{K}^0}(t) = \frac{|q_K/p_K|^2}{4} \\ \times \big[e^{-\Gamma_S t} + e^{-\Gamma_L t} \\ -2e^{-(\Gamma_S+\Gamma_L)t/2}\cos(\Delta m\, t)\big]. \end{aligned} \tag{5.40}$$

Here $\Delta m = m_L - m_S$, and the probability is relative to the initial K^0 preparation, not divided by the survival probability at that time. It contains both oscillation and the two survival envelopes. The amplitude into a common final state f is $A_f g_+ + (q_K/p_K)\bar{A}_f g_-$, with interference between direct decay and decay after mixing. A CP asymmetry requires the relative phases and moduli of these coefficients; the existence of a second-order transfer alone does not fix its value. If the intermediate distinction has been recorded, combine the corresponding conditional histories as in Chapter 3.

§5.6 Strong Decays and Electromagnetic De-excitation

Hadronic resonances and electromagnetic excitations also produce outgoing states through parent–product connections, with different vertices. For an angle-independent spin-averaged squared amplitude, the two-body integral gives

$$\begin{aligned} \Gamma_{A\to 12} &= \frac{p_f}{8\pi M_A^2}\overline{|\mathcal{M}|^2}, \\ p_f &= \frac{\sqrt{\lambda_K(M_A^2, m_1^2, m_2^2)}}{2M_A}, \\ \lambda_K(a,b,c) &= a^2 + b^2 + c^2 \\ &\quad - 2ab - 2ac - 2bc. \end{aligned} \tag{5.41}$$

For example, adopt the effective hadronic vertex $\mathcal{M} = g_{\rho\pi\pi}\epsilon_\rho \cdot (p_+ - p_-)$ for $\rho \to \pi^+\pi^-$. Rest-frame spin averaging gives $\overline{|\mathcal{M}|^2} = 4g_{\rho\pi\pi}^2 p_f^2/3$, hence

$$\begin{aligned} \Gamma_{\rho\to\pi\pi} &= \frac{g_{\rho\pi\pi}^2 p_f^3}{6\pi M_\rho^2} \\ &= \frac{g_{\rho\pi\pi}^2 M_\rho}{48\pi}\left(1 - \frac{4m_\pi^2}{M_\rho^2}\right)^{3/2}. \end{aligned} \tag{5.42}$$

The width contains the hadronic coupling strength and the P-wave threshold suppression p_f^3. The coupling $g_{\rho\pi\pi}$ is determined by projection onto composite-matter matrix elements.

Neutral-pion decay into two photons uses the gauge-invariant pseudoscalar coupling $\mathcal{L}_{\text{int}} = -g_{\pi\gamma\gamma}\pi^0 F_{ab}\widetilde{F}^{ab}/4$. Its amplitude and polarization sum are

$$\begin{aligned} \mathcal{M}_{\gamma\gamma} &= g_{\pi\gamma\gamma}\epsilon^{abcd}\epsilon_{1a}\epsilon_{2b}k_{1c}k_{2d}, \\ \sum |\mathcal{M}_{\gamma\gamma}|^2 &= \frac{1}{2}g_{\pi\gamma\gamma}^2 m_{\pi^0}^4. \end{aligned} \tag{5.43}$$

Dividing by 2! for the two identical photons gives

$$\Gamma_{\pi^0\to\gamma\gamma} = \frac{g_{\pi\gamma\gamma}^2 m_{\pi^0}^3}{64\pi}. \tag{5.44}$$

This hadronic electromagnetic response must be calculated independently. If one additionally adopts the chiral-anomaly matching $g_{\pi\gamma\gamma} = \alpha/(\pi F_\pi)$, the width is $\alpha^2 m_{\pi^0}^3/(64\pi^3 F_\pi^2)$, with convention $f_\pi = \sqrt{2}F_\pi$. Reference [11] discusses the anomaly background and measures the π^0 radiative width. The two-photon channel lacks the helicity-suppression mass factor of charged-pion leptonic decay, allowing the lifetimes to differ by many orders of magnitude.

Photon de-excitation of atoms or nuclei also involves the transition scale. The electric-dipole interaction is $H_{E1} = -\mathbf{d}\cdot\mathbf{E}$. The squared single-photon field amplitude is proportional to ω, and the photon density of states to ω^2. After summing polarizations and integrating directions [12],

$$\Gamma_{E1} = \frac{4\alpha}{3}\omega^3 \overline{\left| \left\langle f \left| \sum_a Q_a \mathbf{r}_a \right| i \right\rangle \right|^2}. \tag{5.45}$$

The charges Q_a are in units of the elementary charge, and the bar includes averaging and summation over magnetic quantum numbers. Transition energy and dipole moment jointly control the lifetime. If the electric-dipole matrix element vanishes, magnetic-dipole, electric-quadrupole, and other allowed terms must be calculated.

§5.7 Open Final States of W and Z Bosons

On-shell W and Z bosons directly form fermion pairs, and their widths are calculated from two-body vertices. For W, first adopt the charged gauge vertex in the same normalization [13]:

$$\mathcal{M}_{W\ell} = \frac{g_W}{2\sqrt{2}}\epsilon_a \bar{u}_\ell \gamma^a (1-\gamma^5) v_\nu. \tag{5.46}$$

Neglecting final-state masses, averaging over the three initial polarizations gives $\overline{|\mathcal{M}_{W\ell}|^2} = g_W^2 M_W^2/3$. The two-body integral yields

$$\Gamma_{W\ell} = \frac{g_W^2 M_W}{48\pi}, \qquad \Gamma_{W,ij} = 3|V_{ij}|^2 \Gamma_{W\ell}. \tag{5.47}$$

If only the up-type flavors u, c are open, all three lepton channels have equal couplings, and the two quark-mixing rows are fully normalized, then $\sum_{i=u,c;j=d,s,b} |V_{ij}|^2 = 2$. The total weight is three leptonic plus six hadronic baseline widths, so

$$\Gamma_W^{(0)} = 9\Gamma_{W\ell}, \qquad B(W\to \ell\nu_\ell)^{(0)} = \frac{1}{9}. \tag{5.48}$$

Nine is a weighted total; individual quark-flavor branches still carry $|V_{ij}|^2$. Mass effects and hadronic radiation must be calculated at the actual W energy and for the outgoing channels.

For Z, adopt the neutral gauge vertex

$$\begin{aligned} \mathcal{M}_{Zf} &= \frac{g_Z}{2}\epsilon_a \bar{u}_f \gamma^a (v_f - a_f\gamma^5) v_{\bar{f}}, \\ v_f &= T_3^f - 2Q_f s_W^2, \qquad a_f = T_3^f. \end{aligned} \tag{5.49}$$

Here s_W^2 is the mixing parameter of the adopted neutral current. Completing the massless two-body integral gives

$$\begin{aligned} \Gamma_{Zf}^{(0)} &= \frac{N_c g_Z^2 M_Z}{48\pi}(v_f^2 + a_f^2), \\ \Gamma_{Z\nu}^{(0)} &= \frac{g_Z^2 M_Z}{96\pi}, \\ \Gamma_{Z,\mathrm{inv}}^{(0)} &= 3\Gamma_{Z\nu}^{(0)}. \end{aligned} \tag{5.50}$$

The last expression assumes three kinematically open light neutrinos with equal neutral-current couplings. The invisible width measures their total share; the absence of direct neutrino absorption by a detector does not mean the products have not formed.

If one further adopts the tree-level relations $g_Z = g_W/\cos\theta_W$, $M_W = M_Z\cos\theta_W$, and $G_F/\sqrt{2} = g_W^2/(8M_W^2)$, these results can be expressed in the usual Fermi-constant form. Independently supplied masses, couplings, and energy scales must first be checked against that common gauge matching. Both on-shell gauge-boson widths scale as g^2M, differing from the $G_F^2M^5$ dependence in the low-energy contact regime.

§5.8 Alpha Decay and Barrier Transmission

The final state of alpha decay contains an alpha particle and the residual nucleus. Even when separation is energetically allowed, nuclear and Coulomb interactions can form a barrier between them. Following the determination of transmission from the propagation equation and flux ratio in matter motion [15], use the relative coordinate r and reduced mass μ_{red}, restoring $\hbar, c$:

$$\begin{aligned} Q_\alpha &= (M_A - M_d - m_\alpha)c^2, \\ \mu_{\mathrm{red}} &= \frac{m_\alpha M_d}{m_\alpha + M_d}, \\ -\frac{\hbar^2}{2\mu_{\mathrm{red}}}u'' + V_{\mathrm{eff}}(r)u &= Q_\alpha u. \end{aligned} \tag{5.51}$$

Outside the nucleus, $V_{\mathrm{eff}}(r) = C/r + \hbar^2\ell(\ell+1)/(2\mu_{\mathrm{red}}r^2)$, where $C = 2Z_de^2/(4\pi\varepsilon_0)$. Between the two turning points, $V_{\mathrm{eff}} > Q_\alpha$, the propagation wave number becomes imaginary, with local spatial attenuation coefficient

$$\kappa(r) = \frac{\sqrt{2\mu_{\mathrm{red}}[V_{\mathrm{eff}}(r) - Q_\alpha]}}{\hbar}. \tag{5.52}$$

Matching the attenuating solution in a slowly varying thick barrier to traveling waves on both sides gives the transmitted flux ratio $\mathcal{T}_\alpha \simeq C_b e^{-2I}$, where $I = \int_{r_1}^{r_2}\kappa(r)dr$, and boundary matching determines C_b. The separation into an alpha preformation factor and a single-particle escape width is discussed in [14]. When the internal nuclear distribution is quasistationary and outgoing products no longer return appreciably, the preformation factor P_α and actual barrier-assault frequency ν_{hit} connect transmission to waiting time:

$$\boxed{\begin{aligned} &\gamma_\alpha \simeq P_\alpha \nu_{\mathrm{hit}} C_b e^{-2I}, \\ &\Gamma_\alpha = \hbar\gamma_\alpha, \qquad t_{1/2}^{(\alpha)} = \frac{\ln 2}{\gamma_\alpha}. \end{aligned}} \tag{5.53}$$

Here $t_{1/2}^{(\alpha)}$ is the partial half-life for alpha decay; it equals the total half-life when alpha decay is the only open channel. Nuclear structure and internal motion determine the preformation factor and assault frequency, respectively.

For the exterior Coulomb potential at $\ell = 0$, integrate from the nuclear surface R to $r_2 = C/Q_\alpha$. With $x = RQ_\alpha/C \in (0,1)$, substituting $r = r_2\sin^2\theta$ evaluates the action integral:

$$I = \frac{C\sqrt{2\mu_{\mathrm{red}}}}{\hbar\sqrt{Q_\alpha}} \times \left[\arccos\sqrt{x} - \sqrt{x(1-x)}\right]. \tag{5.54}$$

For $R \ll r_2$, expansion gives

$$2I \simeq \frac{\pi C\sqrt{2\mu_{\mathrm{red}}}}{\hbar\sqrt{Q_\alpha}} - \frac{4\sqrt{2\mu_{\mathrm{red}}CR}}{\hbar}. \tag{5.55}$$

For a family of nuclei with slowly varying prefactors, the leading term of $\ln t_{1/2}^{(\alpha)}$ therefore varies with $Z_d/\sqrt{Q_\alpha}$, giving a Geiger–Nuttall-type relation. A small energy change is amplified exponentially into a large lifetime change. Angular-momentum barriers, preformation factors, and shell structure further modify the slope corrections and intercept (Figure 5-3).

Spontaneous fission also involves forbidden-region propagation, but uses a collective inertia $B(q)$ and deformation potential $V(q)$ along a nuclear deformation coordinate q, with action $I_{\mathrm{fis}} = \hbar^{-1}\int dq\sqrt{2B(q)[V(q)-E]}$. Its multidimensional paths and nuclear structure require a separate calculation; the reduced mass of the alpha two-body problem cannot simply be reused.

From leptons to nuclei, the actual transfer connection, complete final state, and propagation determine output shares, and the same survival statistics determine lifetime. These specific results now allow comparison of the factors that make an open decay very slow and the conditions that close a channel altogether.

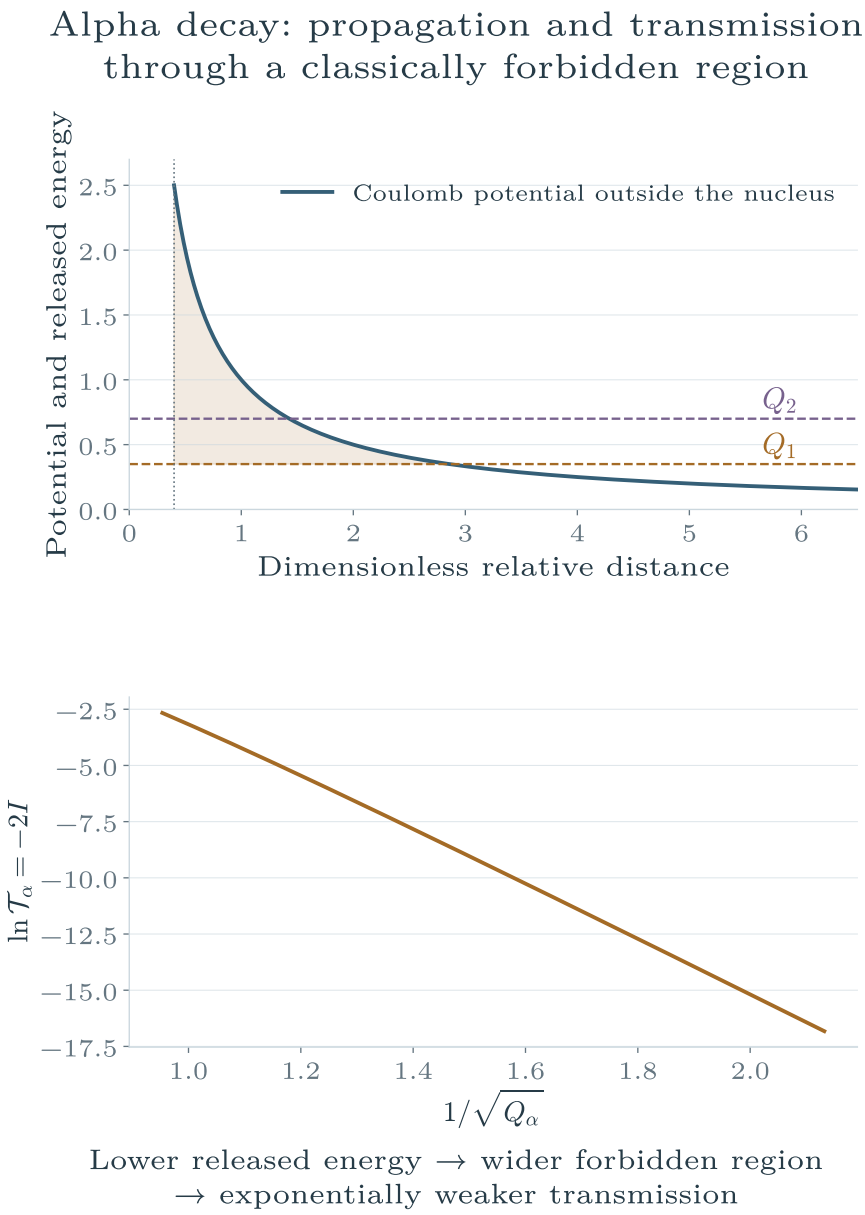


Figure 5-3. Barrier propagation and alpha-decay lifetime. The upper panel shows the exterior Coulomb potential and two available energies; shading marks one corresponding forbidden region. The lower panel plots the logarithm of transmission from the analytic action. Increasing energy narrows the forbidden region and increases transmission. Dimensionless parameters illustrate the exponential mechanism, rather than predict a specific nuclear lifetime.

§6 Lifetime Hierarchies and Stability

Particle lifetimes span many orders of magnitude. The calculations in Chapter 5 attribute these differences to transfer strength, the availability of final states, and product propagation. We now compare how these factors suppress decay rates and establish the distinction between a long lifetime and stability.

§6.1 Open Channels and Lifetime Hierarchies

In the weak-coupling, smooth-outgoing-spectrum regime of Chapter 4, the partial width of a single parent state is $\Gamma_r = 2\pi J_r(E_i)$, with J_r containing both transition amplitudes and available final states. Lifetime hierarchies therefore cannot be ranked by interaction names alone: a stronger interaction can have no open final states, while a weaker interaction can yield a large width when more energy is available.

The muon and tau offer a direct comparison. Both use the same contact vertex, so its coefficient cancels in their width ratio, leaving

$$\frac{\Gamma_\tau}{\Gamma_\mu} = \left(\frac{m_\tau}{m_\mu}\right)^5 \frac{f_e + f_\mu + w_{\text{had}}}{f(m_e^2/m_\mu^2)}. \tag{6.1}$$

A larger parent mass increases the phase space of each open branch, and hadronic outputs add to the total share. Both effects shorten the tau lifetime.

For beta decay, available energy is set by the mass difference between the initial and final composite systems; the total parent nuclear mass cannot replace it. Neglecting electron mass and taking the nuclear matrix element to vary slowly over the relevant range, the phase-space integral contains

$$\int_0^Q E_e^2 (Q - E_e)^2 \, dE_e = \frac{Q^5}{30}. \tag{6.2}$$

A small energy gap can therefore slow an allowed weak transition. Near the electron-mass threshold, the $p_e E_e (E_0 - E_e)^2$ integral of Chapter 5 must be used; the Q^5 law cannot be extrapolated all the way to threshold. Table 6-1 lists the dominant factors in different energy regimes, with all other parameters held fixed in each scaling.

Process and regime	Key factor in the derived width	Source of lifetime differences
Low-energy three-body leptonic decay	$G_F^2 M^5 f(r)$	Contact strength, available energy, final masses
Charged-pion leptonic decay	$m_\ell^2(1 - m_\ell^2/m_\pi^2)^2$	Helicity suppression and two-body phase space
Quark-flavor-changing weak decay	$\lvert V_{ij}\rvert^2$ and hadronic matrix elements	Flavor links and composite-matter response
$\rho \to \pi\pi$	$g_{\rho\pi\pi}^2 p_f^3 / M_\rho^2$	Hadronic coupling and P-wave threshold

Process and regime	Key factor in the derived width	Source of lifetime differences
On-shell W and Z decay	g^2M and open-state weights	Direct two-body output
Electric-dipole de-excitation	$\alpha\omega^3\lvert\mathbf{D}_{fi}\rvert^2$	Energy gap, spatial scale, selection rules
Alpha decay	$\hbar P_\alpha \nu_{\mathrm{hit}} C_b e^{-2I}$	Nuclear preformation and forbidden-region propagation

Table 6-1. Sources of lifetime hierarchies in specific formulas. The rows describe different effective actions and energy regimes; interaction order alone cannot combine them into a universal mass power law.

Successive actions must also be composed according to the actual process. The amplitude of two coherent weak connections contains intermediate energy gaps and different time orderings, and its width requires integration over the complete final state. Neutral-meson mixing additionally retains interference between propagation phases. An intermediate particle that has actually formed and been recorded instead has its own waiting time. For example, two independent exponential waits with rates γ_1, γ_2 give the total waiting-time density by convolution:

$$\begin{aligned} f_T(t) &= \int_0^t \gamma_1 e^{-\gamma_1 s} \gamma_2 e^{-\gamma_2 (t-s)}\, ds \\ &= \frac{\gamma_1\gamma_2}{\gamma_2 - \gamma_1}\left(e^{-\gamma_1 t} - e^{-\gamma_2 t}\right), \\ \mathbb{E}T &= \frac{1}{\gamma_1} + \frac{1}{\gamma_2}. \end{aligned} \tag{6.3}$$

For $\gamma_1 = \gamma_2 = \gamma$, the density tends continuously to $\gamma^2 t e^{-\gamma t}$. Waiting for the cascade endpoint is generally not a single exponential, nor is it obtained by simply multiplying the two decay rates. A lifetime hierarchy must refer to a specified parent particle and first-event definition.

§6.2 Long Lifetime and Stability

An open channel can have a small matrix element or extremely small barrier transmission. If these factors still give a nonzero exit share, the particle retains a nonzero decay probability. Failure to observe decay over a finite time constrains its rate but does not establish that the exit action is exactly zero.

Stability requires checking the reachable range of the complete action. Let P_{st} project onto a physical subspace containing the original particle and its necessary accompanying responses, and let H be the self-adjoint generator on the space considered. If the subspace is closed under the complete action, then

$$\boxed{\begin{aligned} (I - P_{\mathrm{st}}) H P_{\mathrm{st}} &= 0, \\ [H, P_{\mathrm{st}}] &= 0, \\ e^{-iHt/\hbar} P_{\mathrm{st}} &= P_{\mathrm{st}} e^{-iHt/\hbar}. \end{aligned}} \tag{6.4}$$

In the finite reduction, the first relation and self-adjointness directly imply the next two. A continuous representation additionally requires invariance of the relevant operator domain. Including the actual accompanying responses prevents reversible local rearrangements from being mistaken for particle decay. This is a sufficient criterion for closure under the complete dynamics; a vanishing lowest-order on-shell matrix element is a weaker statement.

The record formulation of Chapter 3 gives the same test. For every uninterrupted-survival state reachable from the preparation, suppose all decay outputs satisfy

$$\begin{aligned} &\mathcal{J}_{n,r}(\sigma_{n-1}) = 0 \quad \text{for all } n, r, \\ &S_{n-1} - S_n = \sum_r \operatorname{Tr} \mathcal{J}_{n,r}(\sigma_{n-1}) = 0, \end{aligned} \tag{6.5}$$

Then S_n remains one for a normalized preparation. Conversely, any nonzero decay output from a reachable state reduces the uninterrupted-survival share. Figure 6-1 compares open, suppressed, and closed channels in the same energy picture.

In the adopted particle spectrum, the electron is the lightest negatively charged particle. In isolated vacuum there is no lighter outgoing combination preserving charge, so the electron together with its physical accompanying field has no available decay final state. Heavier charged leptons can preserve charge while becoming an electron and neutrinos; charge conservation alone does not forbid their decay.

For the proton, baryon number is also relevant. Under the baryon-number-conserving actions and existing low-energy spectrum adopted here, no lighter baryonic final state is energetically available. Charge conservation alone is insufficient: the hypothetical channel $p \to e^+ + \pi^0$, for example, preserves charge. Its occurrence depends on a baryon-number-violating action connecting those states, which is not introduced in this paper.

Likewise, the absence of $\mu \to e\gamma$, $\mu \to 3e$, or a flavor-changing neutral-current vertex in a lowest-order list establishes only that the direct amplitude at that order vanishes. If an extended action permits common intermediate states, flavor mixing, or higher-order connections, the corresponding amplitudes must be recalculated. Strict stability, approximate selection rules, and very small nonzero widths must be distinguished within their respective complete actions.

§6.3 Stability and Decay inside Nuclei

A free neutron has an open beta-decay final state, while neutrons in some nuclei persist for long times. A free proton cannot undergo beta-plus decay, while some nuclei containing protons undergo beta-plus decay or electron capture. The difference lies in the complete initial and final nuclear energies and the weak-interaction matrix elements between nuclear states.

Let $M_N(A, Z)$ be the nuclear mass excluding electrons, and initially neglect atomic binding energies and neutrino masses. Write the initial nuclear mass as $M_i = M_N(A, Z)$ and the neighboring nuclear masses as $M_\pm = M_N(A, Z \pm 1)$. For the same mass number A, the released energies are

$$\begin{aligned} Q_{\beta^-} &= (M_i - M_+ - m_e)c^2, \\ Q_{\beta^+} &= (M_i - M_- - m_e)c^2, \\ Q_{\mathrm{EC}} &= (M_i + m_e - M_-)c^2. \end{aligned} \tag{6.6}$$

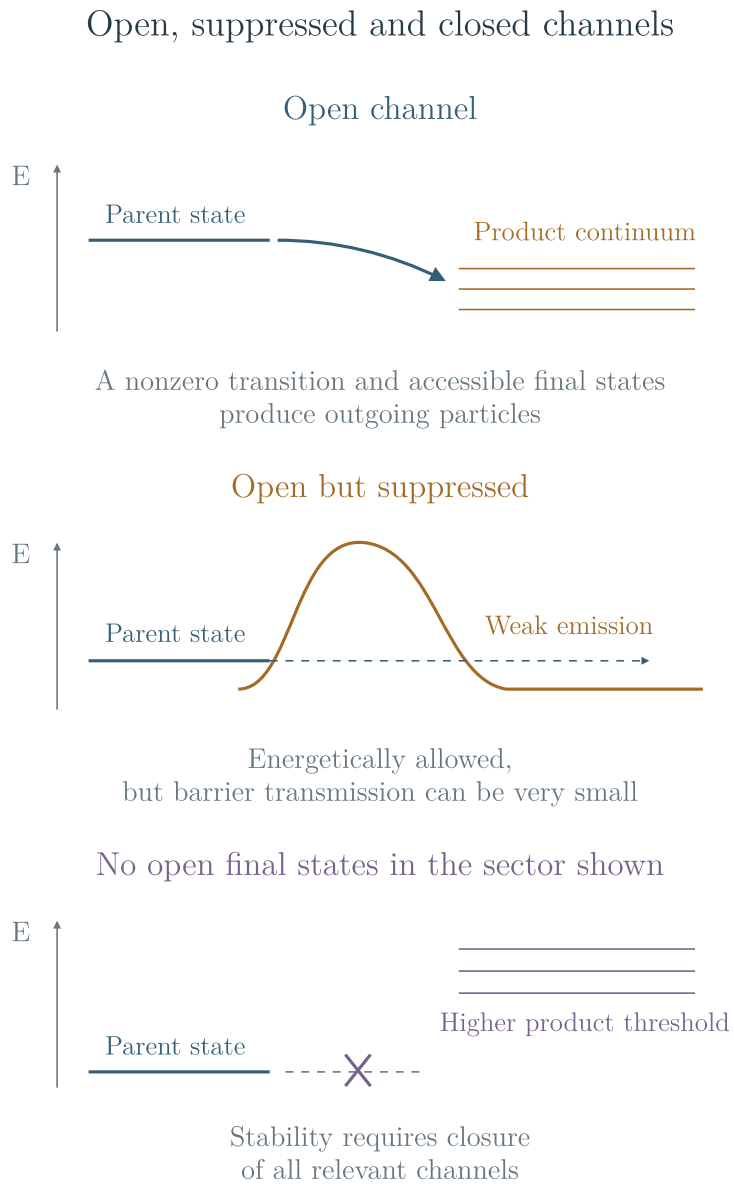


Figure 6-1. Open channels, long lifetimes, and stability. In the upper panel the product threshold lies below the parent and a nonzero transition can form outgoing products. In the middle panel energy permits decay, but a barrier makes transmission small. The lower panel has no reachable outgoing state under the action and conserved sector shown. The third situation must hold for all relevant channels to establish stability.

The electron-capture expression uses an initial nucleus and an electron at rest. Bound-electron energies, atomic vacancies, and daughter-nucleus excitation must enter the energy balance of the actual process. With neutral-atom masses instead, beta-plus decay has the familiar two-electron-mass threshold. The two mass conventions must not be mixed.

A negative Q closes the corresponding process for the stated initial preparation. For $Q > 0$, nuclear spin, parity, and transition matrix elements must still be checked. Allowed transitions use the nuclear response of Chapter 5; forbidden transitions are governed by higher multipoles and their kinematic factors. Daughter excitation reduces the available energy. Different excited states form separate branches and can subsequently emit photons.

Nuclear binding therefore changes both thresholds and responses. It can close the exit available to a free neutron or open a composite-system channel unavailable to a free proton. Nuclear stability also requires checking all actually reachable channels, including beta decay, alpha decay, and fission. Stability against one decay class does not establish stability against every breakup process.

§6.4 Direction of Spontaneous Decay

Spontaneous decay starts from a prepared parent particle and specified surroundings. The products propagate outward and leave correlations in surrounding degrees of freedom, while amplitudes maintaining the original particle and amplitudes of products evolve under the same joint dynamics. The initial state already contains releasable energy, so transfer can occur through an open channel without incoming thermal excitation.

Joint evolution remains reversible. For any initial and final states and unitary action $U(t)$,

$$|\langle f|U(t)|i\rangle|^2 = |\langle i|U^\dagger(t)|f\rangle|^2. \tag{6.7}$$

The right side uses the inverse action and the corresponding final-state preparation. It does not imply that an ordinary environment automatically reproduces the reverse process. If f includes the motions and phases of several products and records already dispersed into the environment, complete reversal requires the corresponding joint incoming state and control of its correlations. Ordinary preparations usually provide one localized parent and let its products leave, so repeated experiments exhibit a direction of persistent exit.

An inverse reaction can still occur with suitable incoming conditions; a common final state can, for example, re-form the corresponding resonance. Its rate is calculated from the new incoming state and the same connection. The resulting particle begins a new survival history.

§7 Conclusion

The decay of matter is a transfer into products under the same dynamics that maintains the original particle. The complete action supports its persistence while connecting it to multiparticle outputs with the same total conserved quantities. Local exchange, product propagation, and return jointly determine the distribution between these components. Once actual outputs preserve distinctions, uninterrupted survival and first decay obey a probability balance that can be checked round by round:

$$\boxed{S_{n-1} - S_n = \sum_r P(n, r).} \tag{7.1}$$

Here $P(n, r)$ is the probability of first decay into branch r in round n. Accumulating this balance with the complete duration of each round gives waiting time; retaining product states gives spectra, angular distributions, branches, and subsequent cascades. For a single narrow resonance formed under weak coupling and a smooth outgoing spectrum, the same parent–product coupling gives

$$\boxed{\begin{aligned} &\Gamma_r = 2\pi J_r(E_i), \qquad \Gamma = \sum_r \Gamma_r, \\ &B_r = \frac{\Gamma_r}{\Gamma}, \qquad \overline{T}_{\text{cont}} = \frac{\hbar}{\Gamma}. \end{aligned}} \tag{7.2}$$

The decay of survival probability in time is thereby related to the resonance width in energy (Figure 7-1). The exponential law applies within the relevant effective regime. Shorter- and longer-time behavior, as well as finite-round time assignment, remain governed by the complete survival action.

The wide range of lifetimes has traceable structural sources. Energy sharing in contact three-body leptonic decay gives both a width proportional to the fifth power of the parent mass and a continuous spectrum. The chiral contraction in charged-pion decay leaves a lepton-mass factor, suppressing the lighter electron branch. Nuclear gaps, angular momentum, and composite responses alter the available final states, while barrier propagation amplifies small energy differences into exponential lifetime differences. Common final states retain coherent mixing for neutral mesons; a cascade with intermediate records accumulates successive conditional outputs and waits. Each result is expressed through explicit transition amplitudes, final-state integrals, and propagation equations.

This paper connects the dynamics of the Bost–Connes state space to these physical calculations. Leptonic lifetimes, hadronic branches, and nucleon responses use the effective vertices, strength matching, and return models specified in the text. Actual hadronic matrix elements, nuclear structure, and radiative terms remain to be evaluated further. Stability is judged by closure of all relevant exit actions; a long lifetime means that a nonzero exit is suppressed. Why matter persists, when it transforms, and what spectra and records it leaves can thus be described and calculated within the same dynamical process.

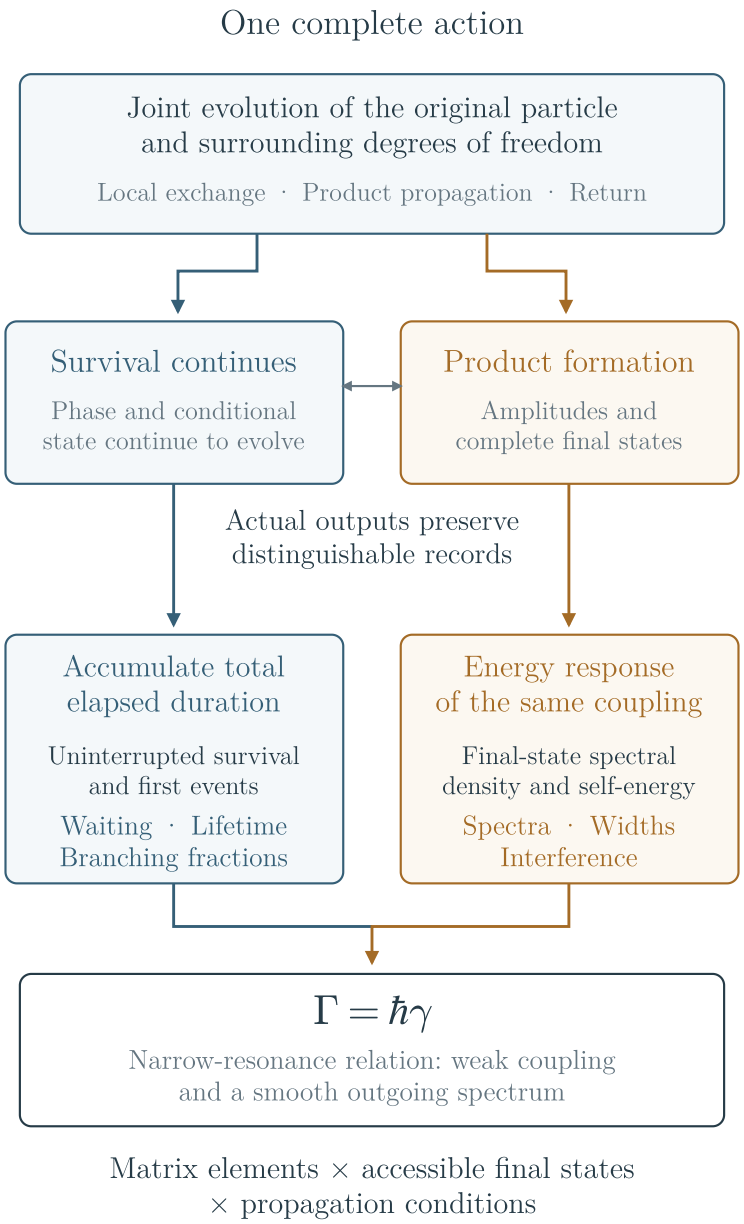


Figure 7-1. Survival and decay under one complete action. Actual transfer connections give product amplitudes; outflow and records establish distinguishable events. Weighting the full elapsed durations by survival probabilities gives the lifetime, while the energy response of the same coupling gives the spectral width. Matrix elements, open final states, and propagation jointly determine the physical results.

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